

Chapter 10

Preserve Environment via Stack

Z. Gu

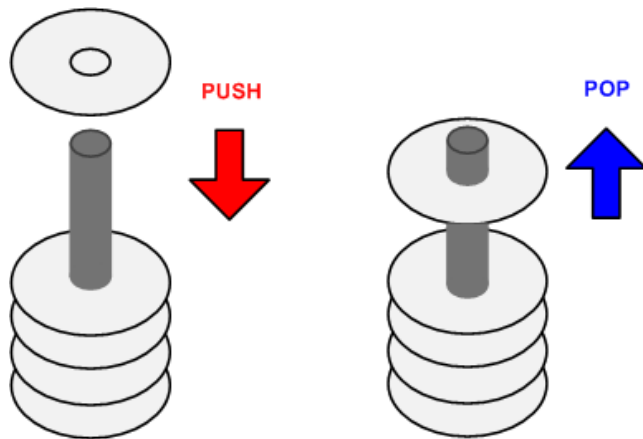
Fall 2025

Overview

- ▶ How to call a subroutine?
- ▶ How to return the control back to the caller?
- ▶ How to pass arguments into a subroutine?
- ▶ How to return a value in a subroutine?
- ▶ How to preserve the running environment for the caller?

Stack

- ▶ A **Last-In-First-Out** memory model
- ▶ Only allow to access the most recently added item
 - ▶ Also called the top of the stack
- ▶ Key operations:
 - ▶ **push** (add item to stack)
 - ▶ **pop** (remove top item from stack)
- ▶ Example: Tower of Hanoi
 - ▶ Only one disk may be moved at a time.
 - ▶ Each move consists of taking the upper disk from one of the rods and sliding it onto another rod, on top of the other disks that may already be present on that rod.
 - ▶ **No disk may be placed on top of a smaller disk.**

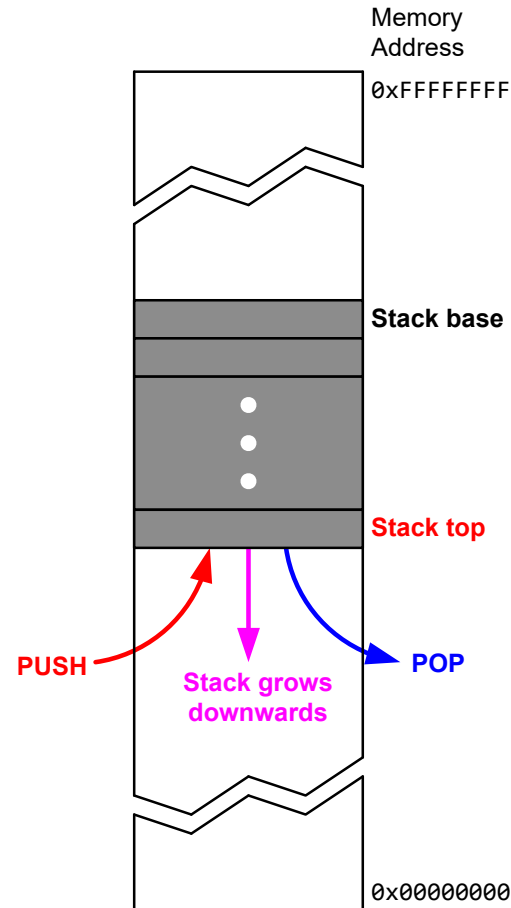


http://en.wikipedia.org/wiki/File:Tower_of_Hanoi_4.gif

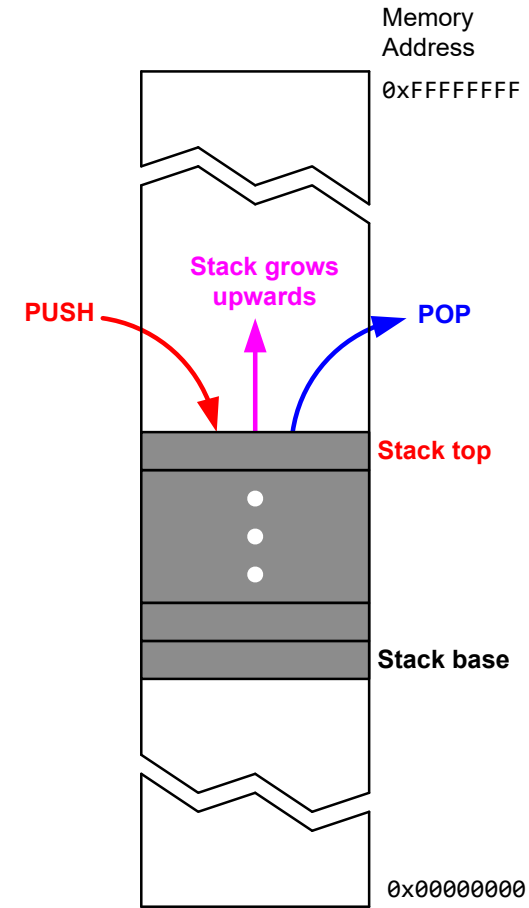
Typical Usage of Stack

- ▶ Why need stack?
 - ▶ Saving the original contents of processor's registers at the beginning a subroutine (Contents are restored at the end of a subroutine)
 - ▶ Storing local variables in a subroutine
 - ▶ Passing extra arguments to a subroutine
 - ▶ Saving processor's registers upon an interrupt

Stack Growth Convention: Ascending *vs* Descending



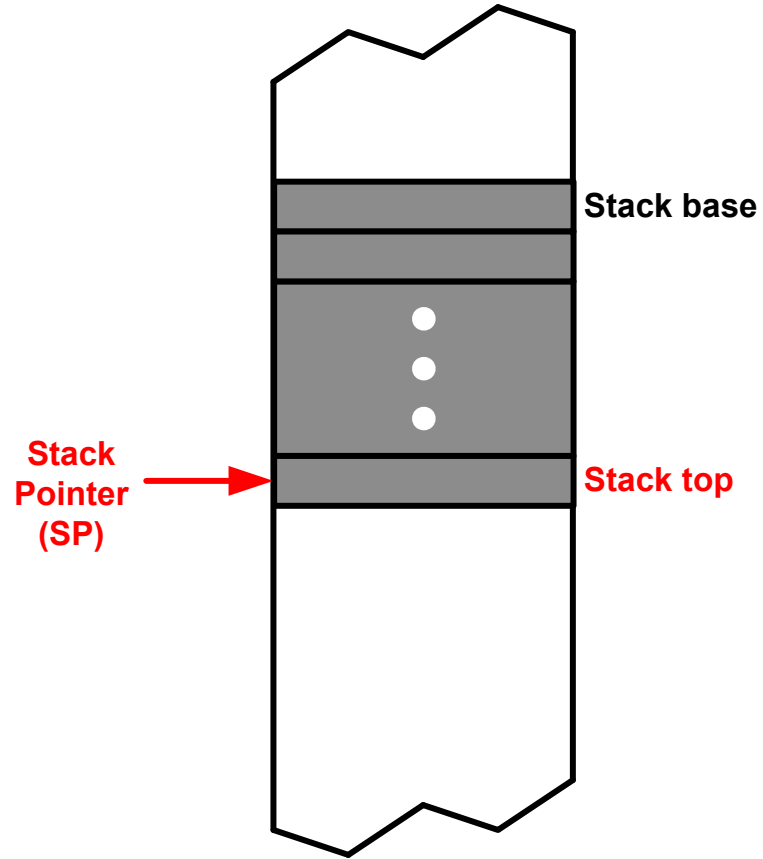
Descending stack: Stack grows towards low memory address



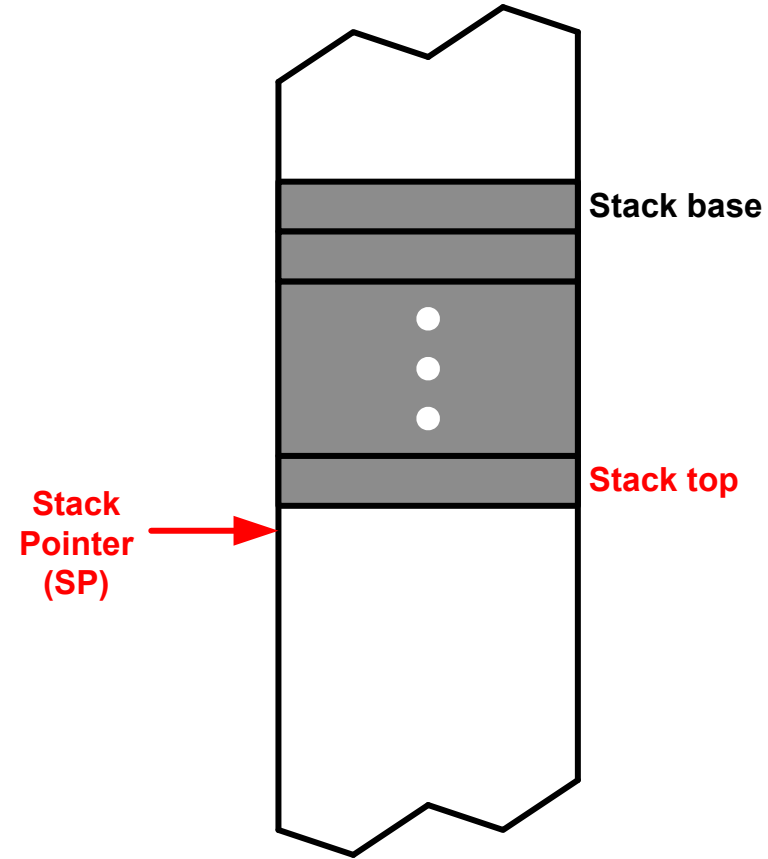
Ascending stack: Stack grows towards high memory address

Stack Growth Convention:

Full *vs* Empty



Full stack: SP points to the last item pushed onto the stack



Empty stack: SP points to the next free space on the stack

Cortex-M Stack

- ▶ Cortex-M uses **full descending stack**

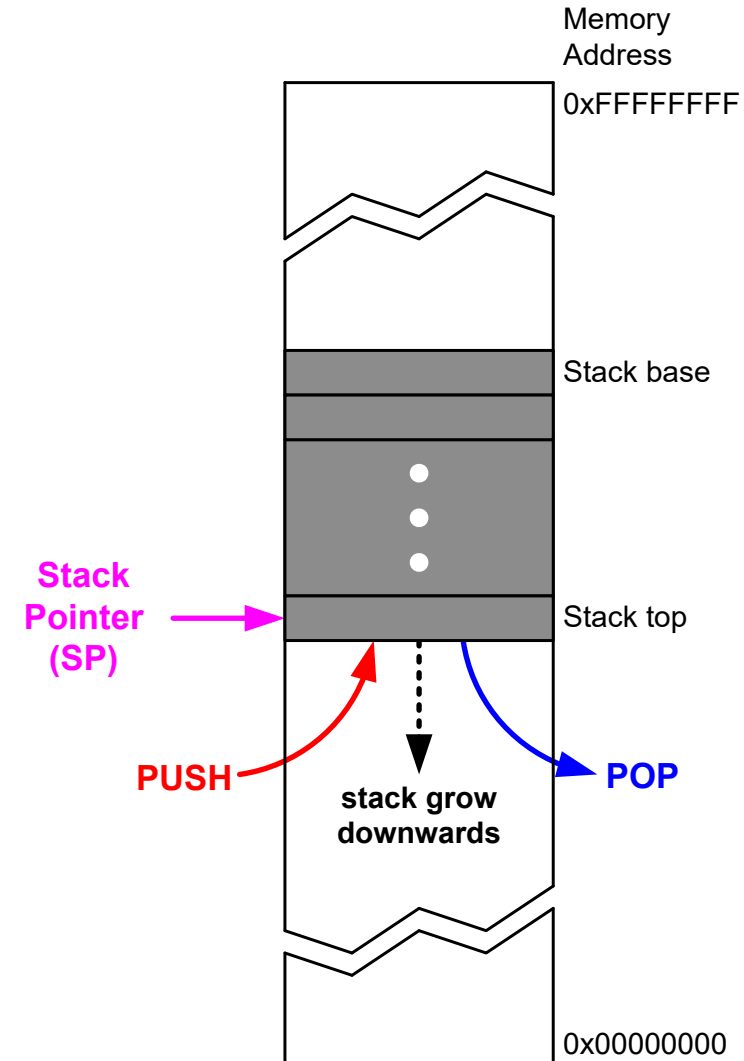
- ▶ Example:

PUSH/POP {r0, r6, r3}

- ▶ Stack pointer (SP, aka R13)

- ▶ decremented on **PUSH**
 - ▶ $SP = SP - 4 * \# \text{ of registers}$
- ▶ incremented on **POP**
 - ▶ $SP = SP + 4 * \# \text{ of registers}$

- ▶ SP starts at **0x20000200** for STM32-Discovery by default (can be changed in startup.s)



Addressing Modes for Load/Store Multiple Registers

STMxx rn{!}, {register_list}

LDMxx rn{!}, {register_list}

- xx = IA, IB, DA, or DB

Addressing Modes	Description	Instructions
IA	Increment A fter	STMIA, LDMIA
IB	Increment B efore	STMIB, LDMIB
DA	Decrement A fter	STMDA, LDMDA
DB	Decrement B efore	STMDB, LDMDB

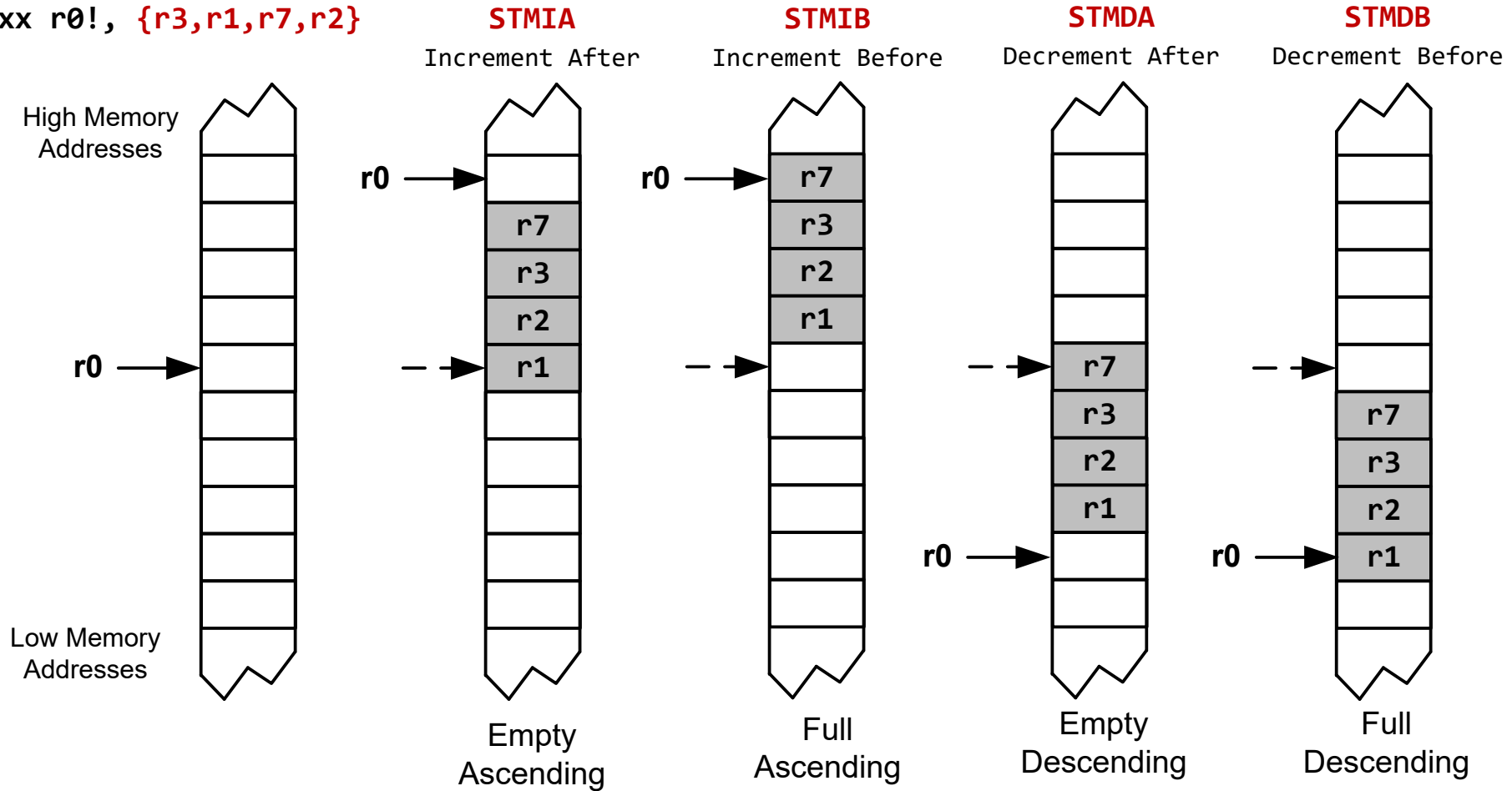
- **IA**: address is incremented by 4 after a word is loaded or stored.
- **IB**: address is incremented by 4 before a word is loaded or stored.
- **DA**: address is decremented by 4 after a word is loaded or stored.
- **DB**: address is decremented by 4 before a word is loaded or stored.

Load/Store Multiple Registers

- ▶ The following are synonyms.
 - ▶ **STM** = **STMIA** (Increment After) = **STM_{EA}** (Empty Ascending)
 - ▶ **LDM** = **LDMIA** (Increment After) = **LDM_{FD}** (Full Descending)
- ▶ The order in which registers are listed does not matter
 - ▶ For STM/LDM, the lowest-numbered register is stored/loaded at the lowest memory address.

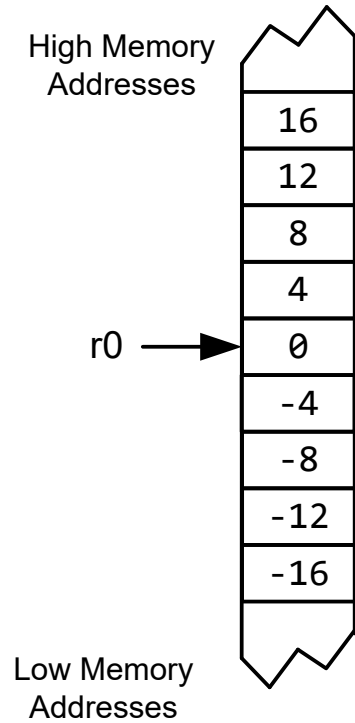
Store Multiple Registers

STMxx r0!, {r3,r1,r7,r2}



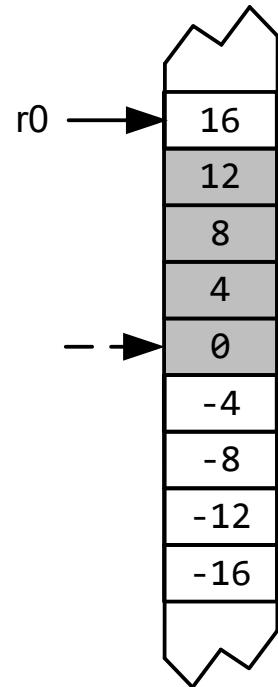
Load Multiple Registers

LDMxx r0!, {r3,r1,r7,r2}



LDMIA

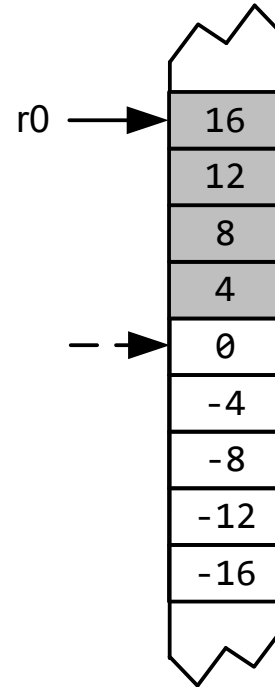
Increment After



r1 = 0
r2 = 4
r3 = 8
r7 = 12

LDMIB

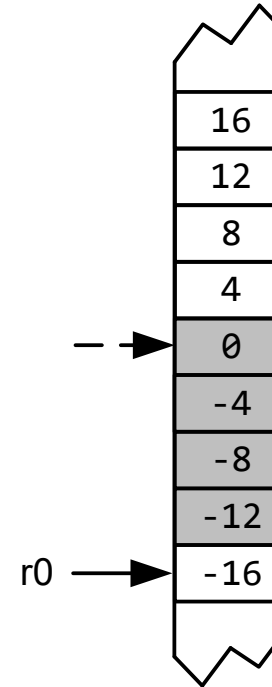
Increment Before



r1 = 4
r2 = 8
r3 = 12
r7 = 16

LDMDA

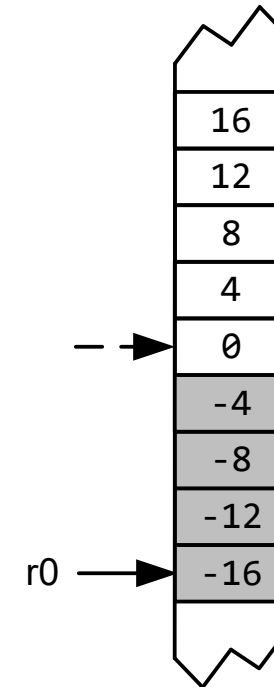
Decrement After



r1 = -12
r2 = -8
r3 = -4
r7 = -0

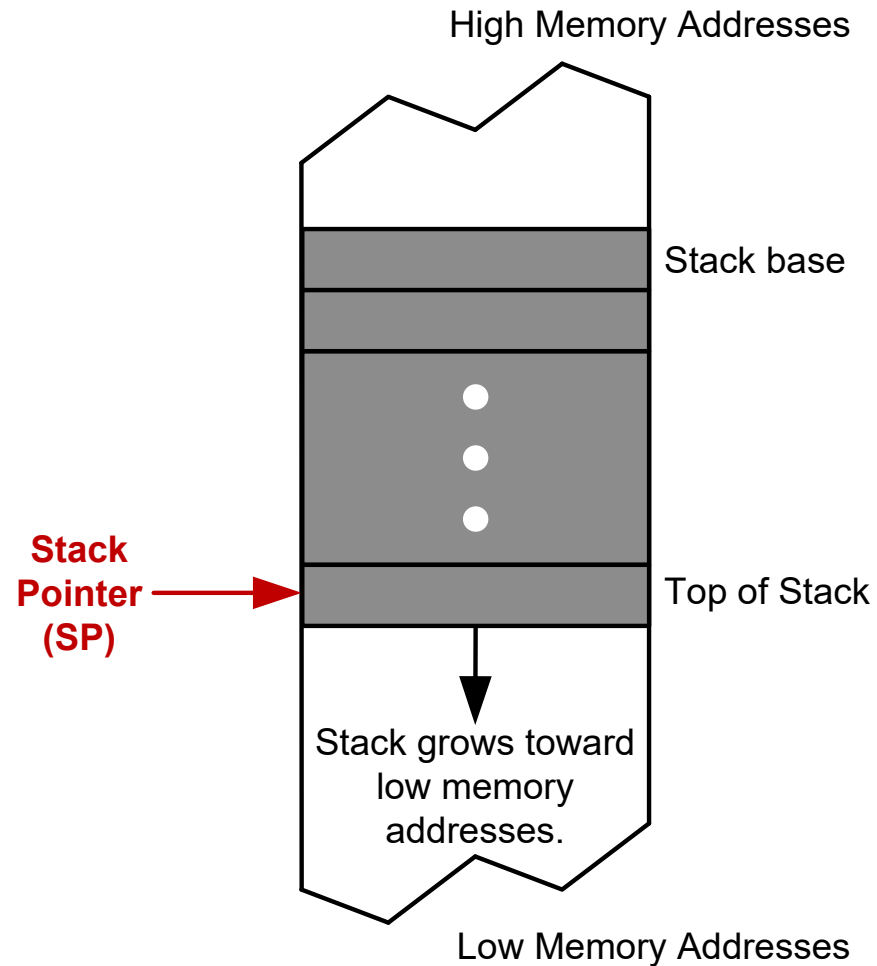
LDMDB

Decrement Before



r1 = -16
r2 = -12
r3 = -8
r7 = -4

Full Descending Stack



PUSH {register_list}
equivalent to:
STMDB SP!, {register_list}

DB: Decrement Before

POP {register_list}
equivalent to:
LDMIA SP!, {register_list}

IA: Increment After

Stack Implementation

Stack Name	Push		Pop	
	Equivalent	Alternative	Equivalent	Alternative
Full Descending(FD)	STMFD SP!,list	STMDB SP!,list	LDMFD SP!,list	LDMIA SP!,list
Empty Descending(ED)	STMED SP!,list	STMDA SP!,list	LDMED SP!,list	LDMIB SP!,list
Full Ascending(FA)	STMFA SP!,list	STMIB SP!,list	LDMFA SP!,list	LDMDA SP!,list
Empty Ascending(EA)	STMEA SP!,list	STMIA SP!,list	LDMEA SP!,list	LDMDB SP!,list

Stack

PUSH {Rd} == STMDB SP!, {Rd} == STMFD SP!, {Rd}

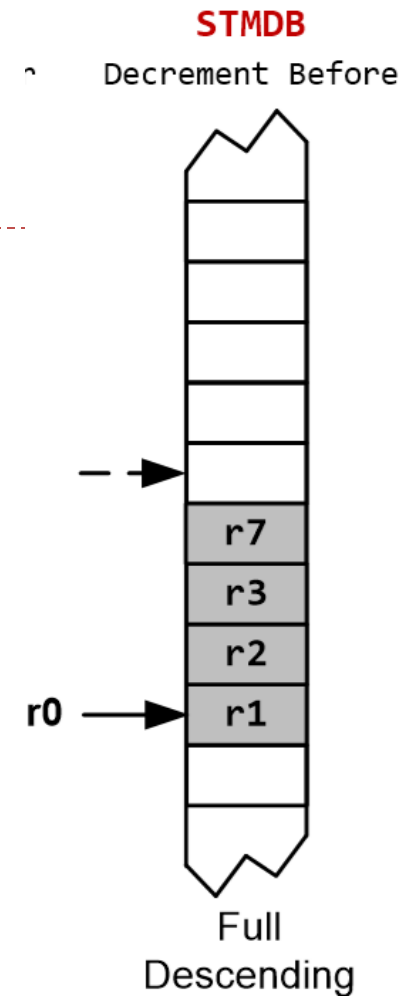
- ▶ $SP = SP - 4 \rightarrow$ descending stack
- ▶ $(*SP) = Rd \rightarrow$ full stack

Push multiple registers

PUSH {r1, r2, r3, r7} $\longleftrightarrow$ *They are equivalent.* **PUSH {r7, r2, r3, r1}** $\longleftrightarrow$

PUSH {r7}
PUSH {r3}
PUSH {r2}
PUSH {r1}

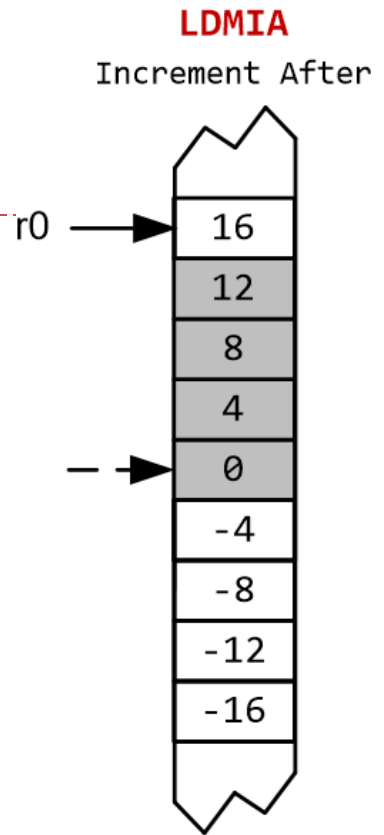
- The order in which registers listed in the register list does not matter.
- When pushing multiple registers, these registers are automatically **sorted by name** and **the lowest-numbered register** is stored to the lowest memory address, i.e. **is stored last**.



Stack

POP {Rd} == LDMIA SP!, {Rd} == LDMFD SP!, {Rd}

- ▶ $Rd = (*SP) \rightarrow$ full stack
- ▶ $SP = SP + 4 \rightarrow$ Stack shrinks



Pop multiple registers

They are equivalent.

POP {r1, r2, r3, r7} $\longleftrightarrow$ POP {r7, r3, r2, r1} $\longleftrightarrow$

**POP {r1}
POP {r2}
POP {r3}
POP {r7}**

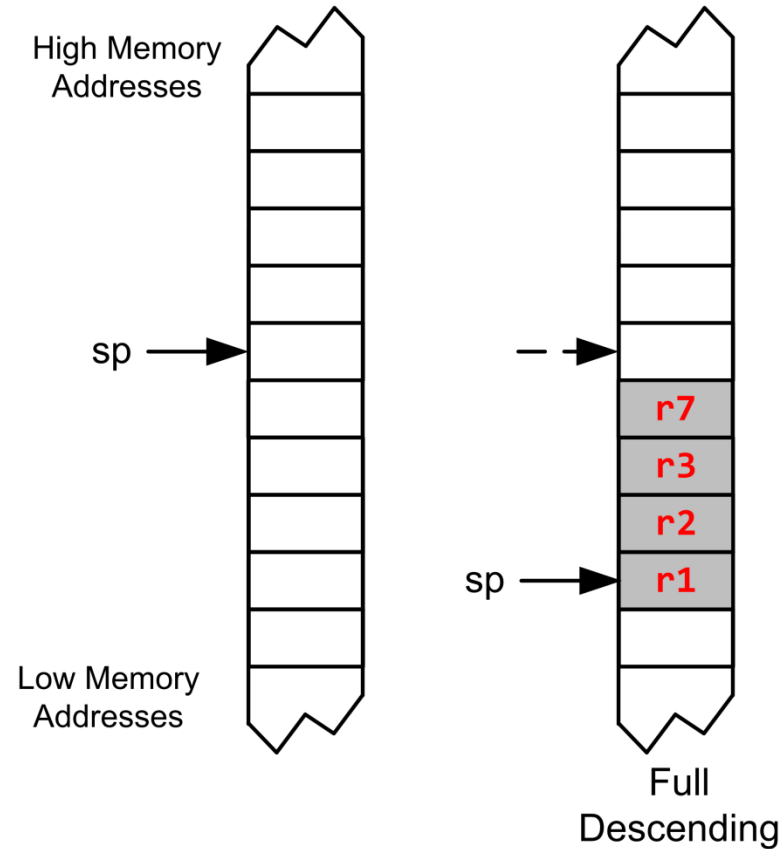
**r1 = 0
r2 = 4
r3 = 8
r7 = 12**

- The order in which registers listed in the register list does not matter.
- When popping multiple registers, these registers are automatically **sorted by name** and **the lowest-numbered register** is loaded from the lowest memory address, i.e. **is loaded first**.

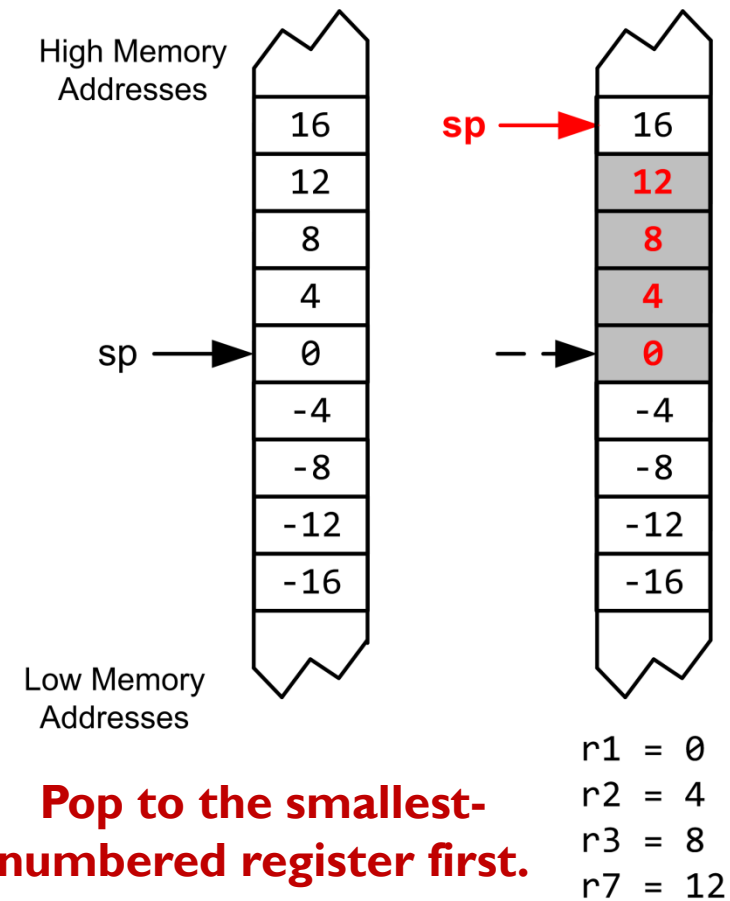
Stack Recap

Largest-numbered register is pushed first but popped last.

PUSH {r3, r1, r7, r2}



POP {r3, r1, r7, r2}



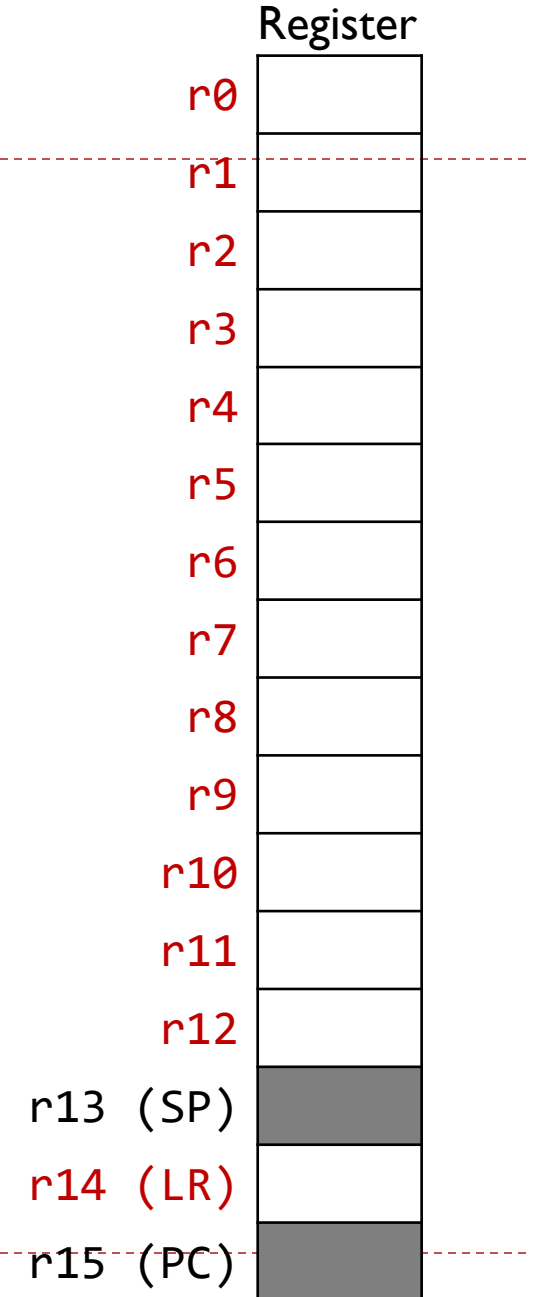
Pop to the smallest-numbered register first.

Cortex-M Stack

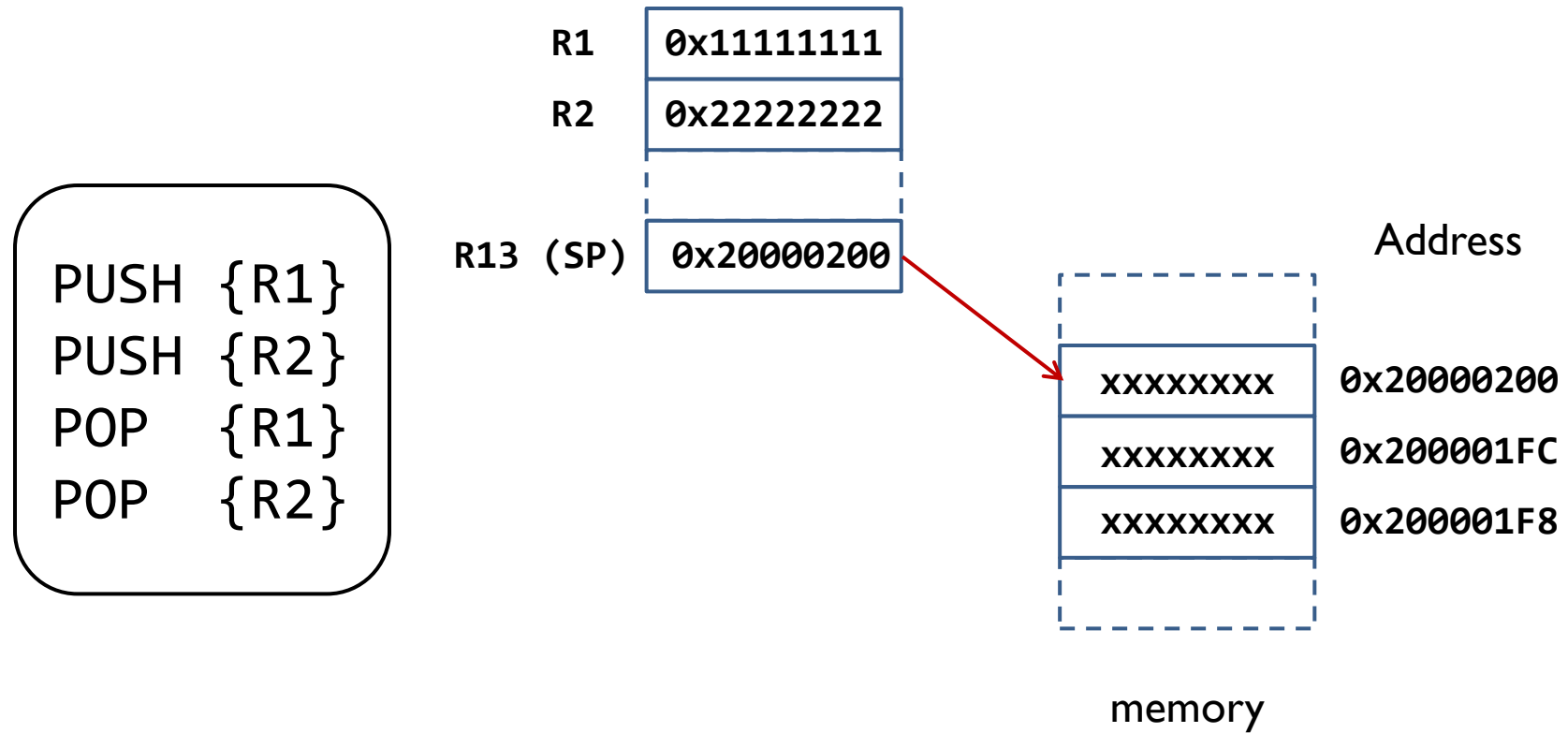
- ▶ **PUSH {register list}**
- ▶ **POP {register list}**
 - ▶ Eligible registers for PUSH: r0 – r12, LR
 - ▶ Eligible registers for POP: r0 – r12, LR, PC
 - ▶ Order specified in the list does not matter
 - ▶ Largest-numbered register is always pushed first and popped last
 - ▶ Lowest-numbered register stored at lowest memory address

Instruction format for PUSH:

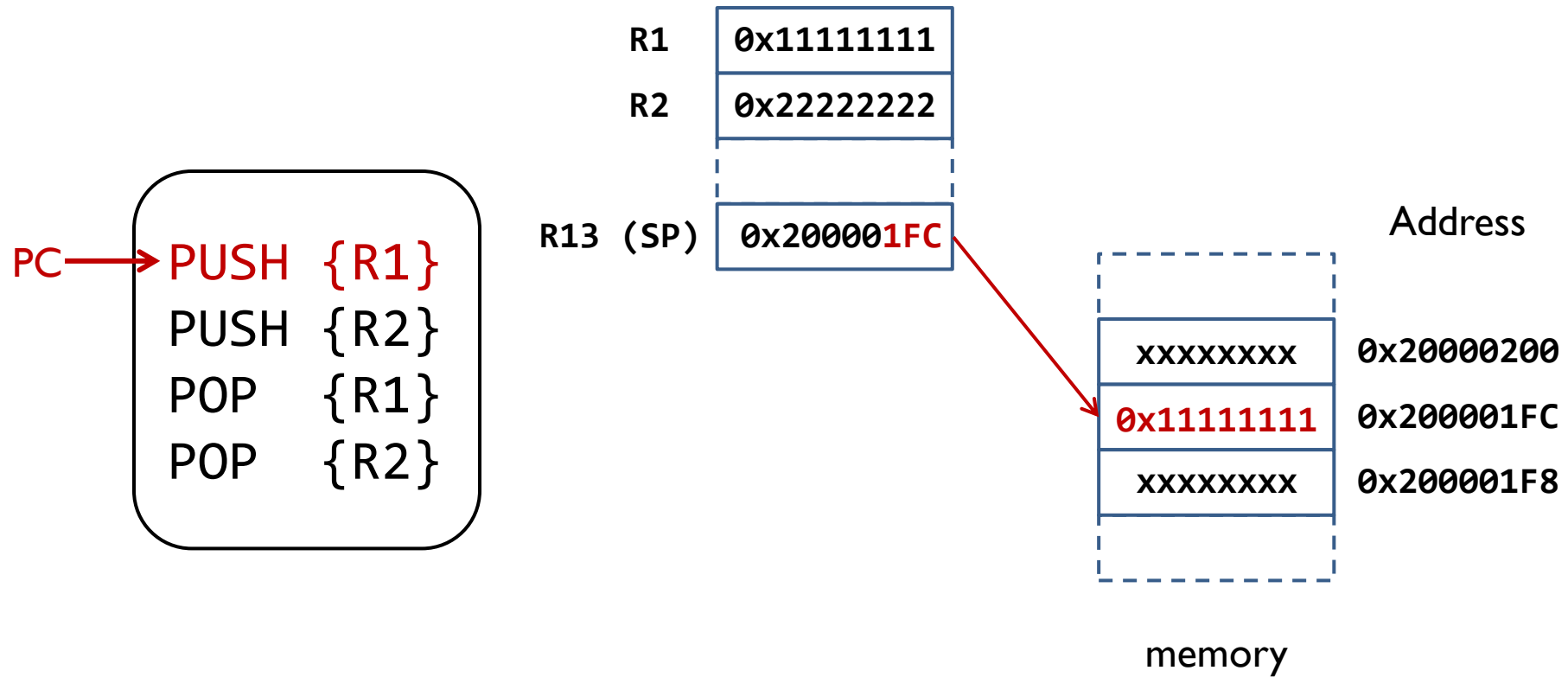
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	0	1	0	0	1	0	0	1	0	1	1	0	1	(0)	M	(0)	register_list												



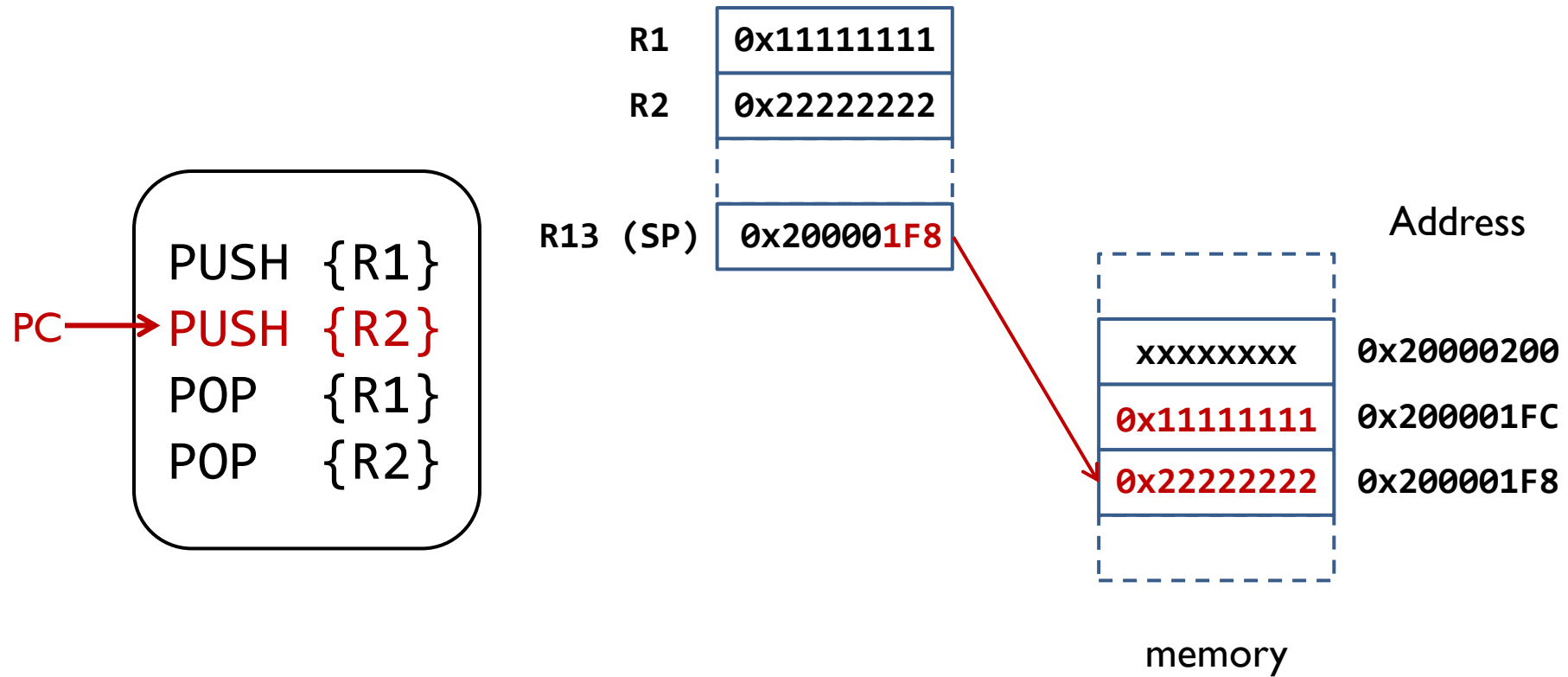
Example: swap R1 & R2



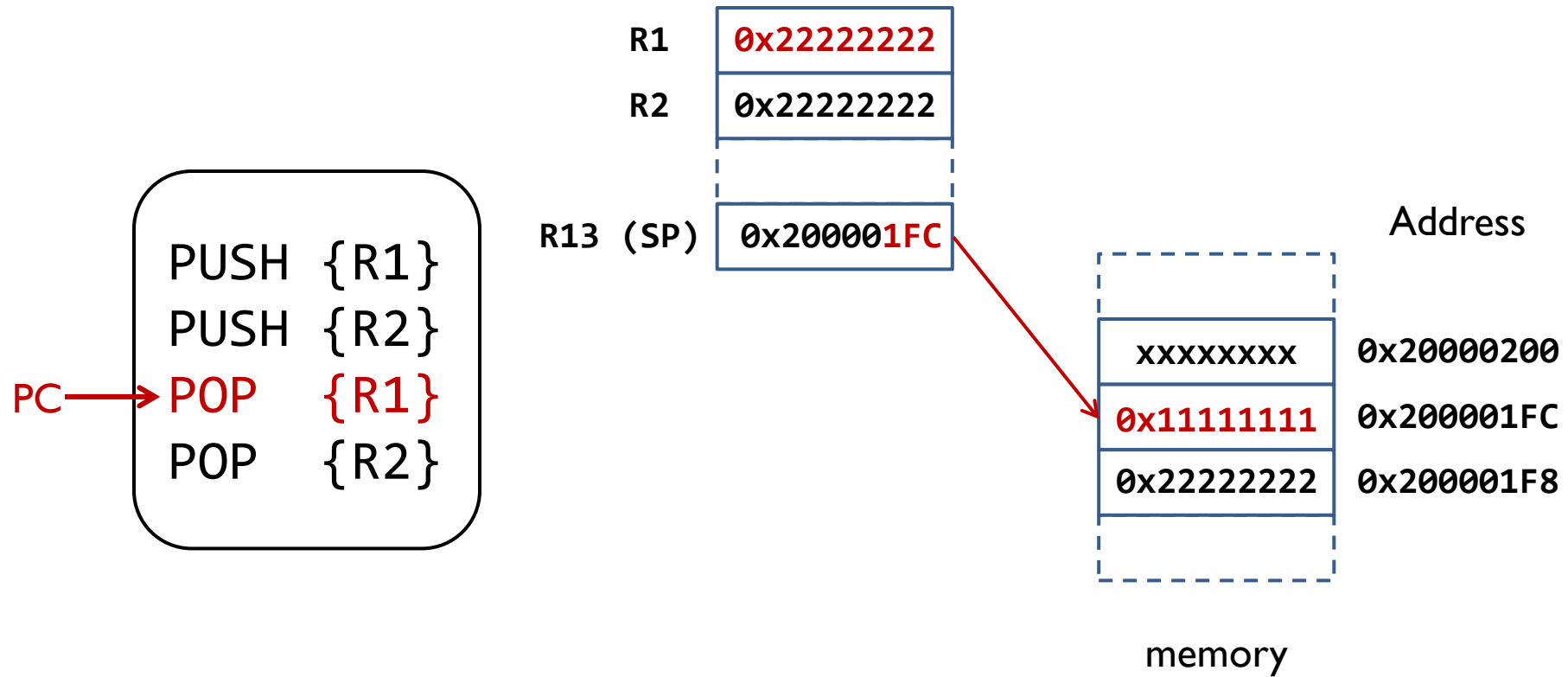
Example: swap R1 & R2



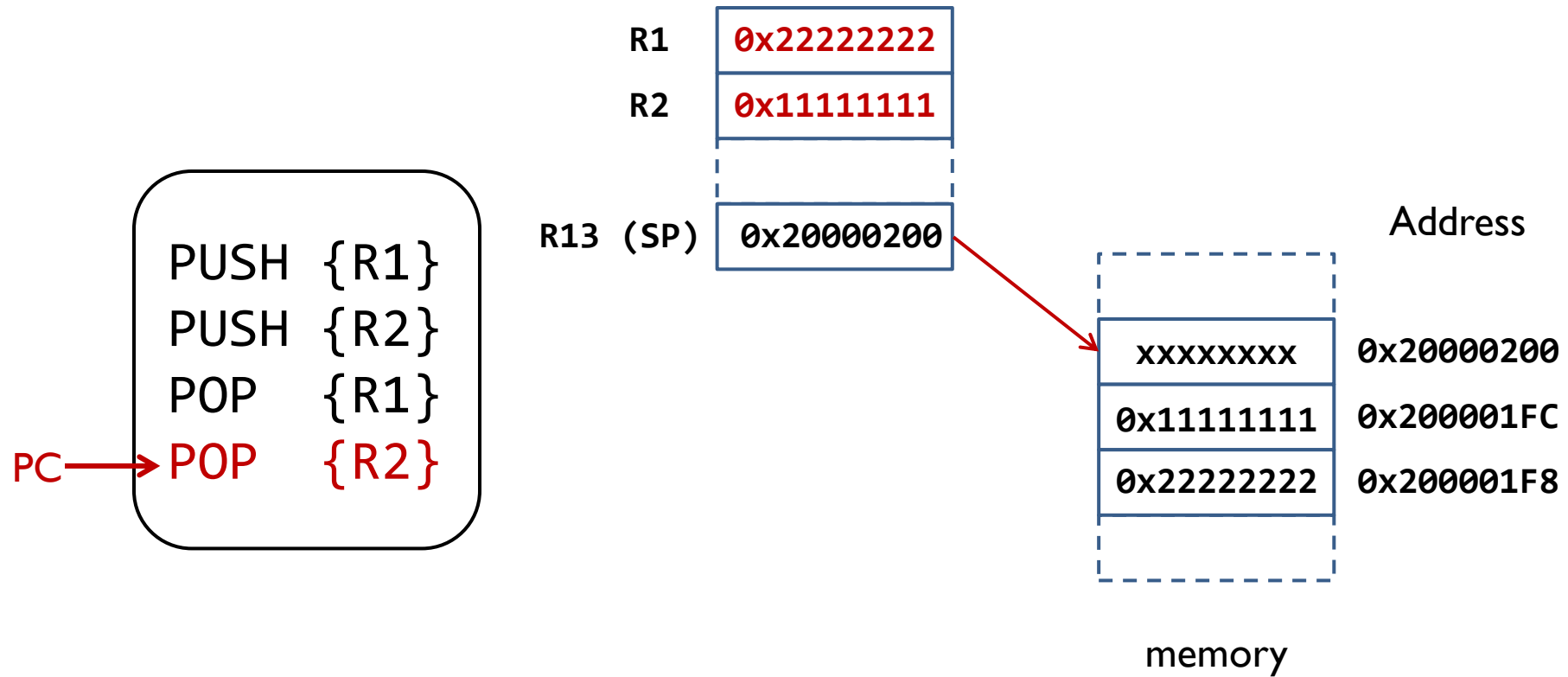
Example: swap R1 & R2



Example: swap R1 & R2



Example: swap R1 & R2



Quiz

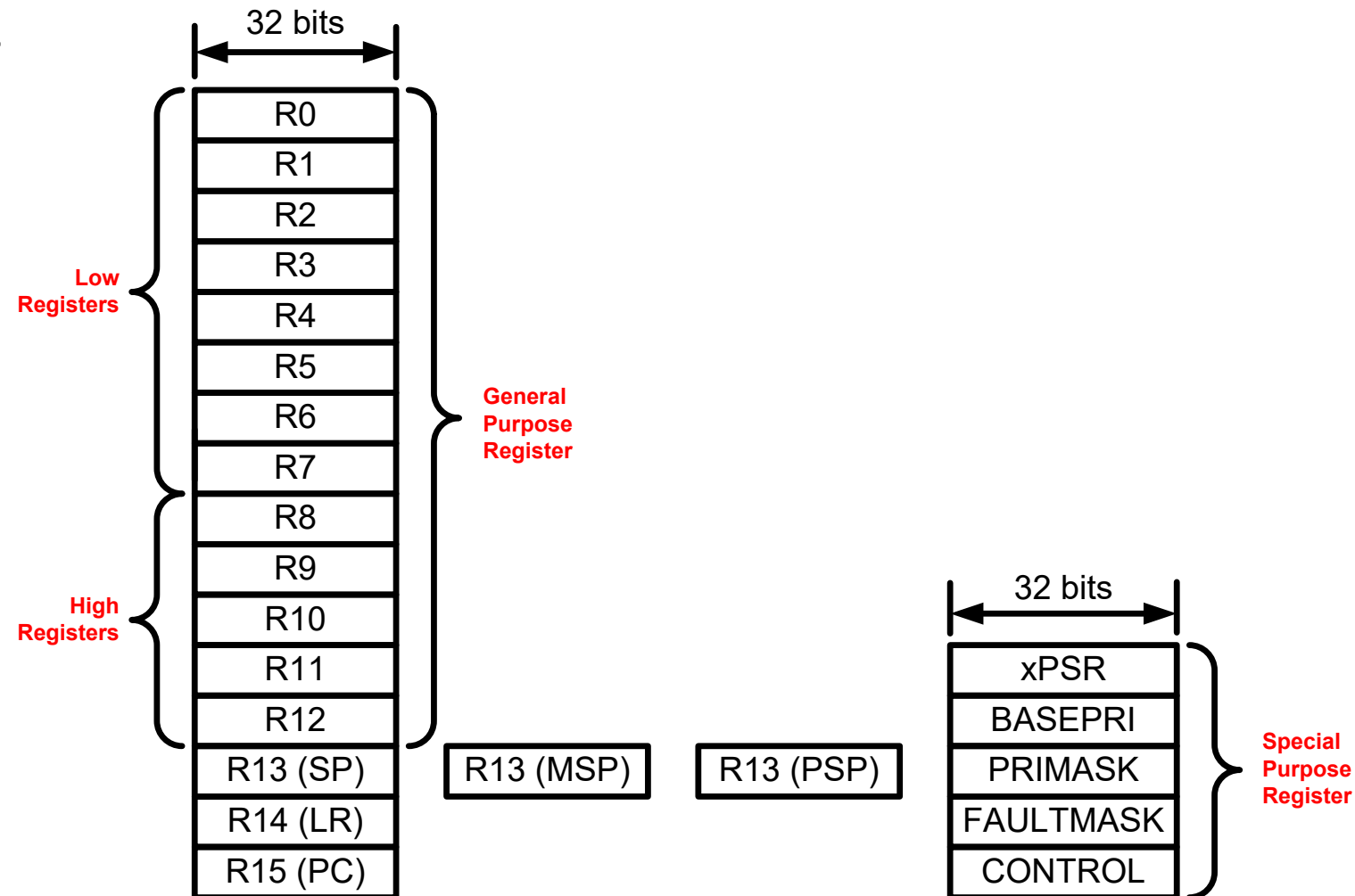
- ▶ Are the values of R1 and R2 swapped?
- ▶ `PUSH {R1, R2}; POP {R2, R1}`
- ▶ Or
- ▶ `PUSH {R1, R2}; POP {R2}; POP {R1}`
- ▶ Or
- ▶ `PUSH {R1}; PUSH {R2}; POP {R1, R2}`

Quiz ANS

- ▶ Are the values of R1 and R2 swapped?
- ▶ `PUSH {R1, R2}; POP {R2, R1}`
- ▶ Or
- ▶ `PUSH {R1, R2}; POP {R2}; POP {R1}`
- ▶ Or
- ▶ `PUSH {R1}; PUSH {R2}; POP {R1, R2}`
- ▶ `PUSH {R1, R2}; POP {R2, R1}`
 - ▶ is equivalent to
 - ▶ `PUSH {R2}; PUSH {R1}; POP{R1}; POP{R2}`
 - ▶ Values of R1 and R2 are unchanged
- ▶ `PUSH {R1, R2}; POP {R2}; POP {R1}`
 - ▶ is equivalent to
 - ▶ `PUSH {R2}; PUSH {R1}; POP{R2}; POP{R1}`
 - ▶ Values of R1 and R2 are swapped
- ▶ `PUSH {R1}; PUSH {R2}; POP {R1, R2}`
 - ▶ is equivalent to
 - ▶ `PUSH {R1}; PUSH {R2}; POP{R1}; POP{R2}`
 - ▶ Values of R1 and R2 are swapped

Subroutine

- ▶ A subroutines, also called a function or a procedure,
 - ▶ single-entry, single-exit
 - ▶ Return to caller after it exits
- ▶ When a subroutine is called, the **Link Register** (LR) holds the return address of the current subroutine call, i.e., memory address of the next instruction to be executed after the subroutine exits.



Calling a Subroutine

Caller: **BL *label*** (Branch and Link)

- ▶ Step 1: $LR = PC + 4$
- ▶ Step 2: $PC = label$
 - ▶ *label* is name of subroutine
 - ▶ Compiler translates label to memory address
 - ▶ After call, LR holds return address (the instruction following the call)

Callee: **BX LR** (Branch and Exchange)
at end of procedure

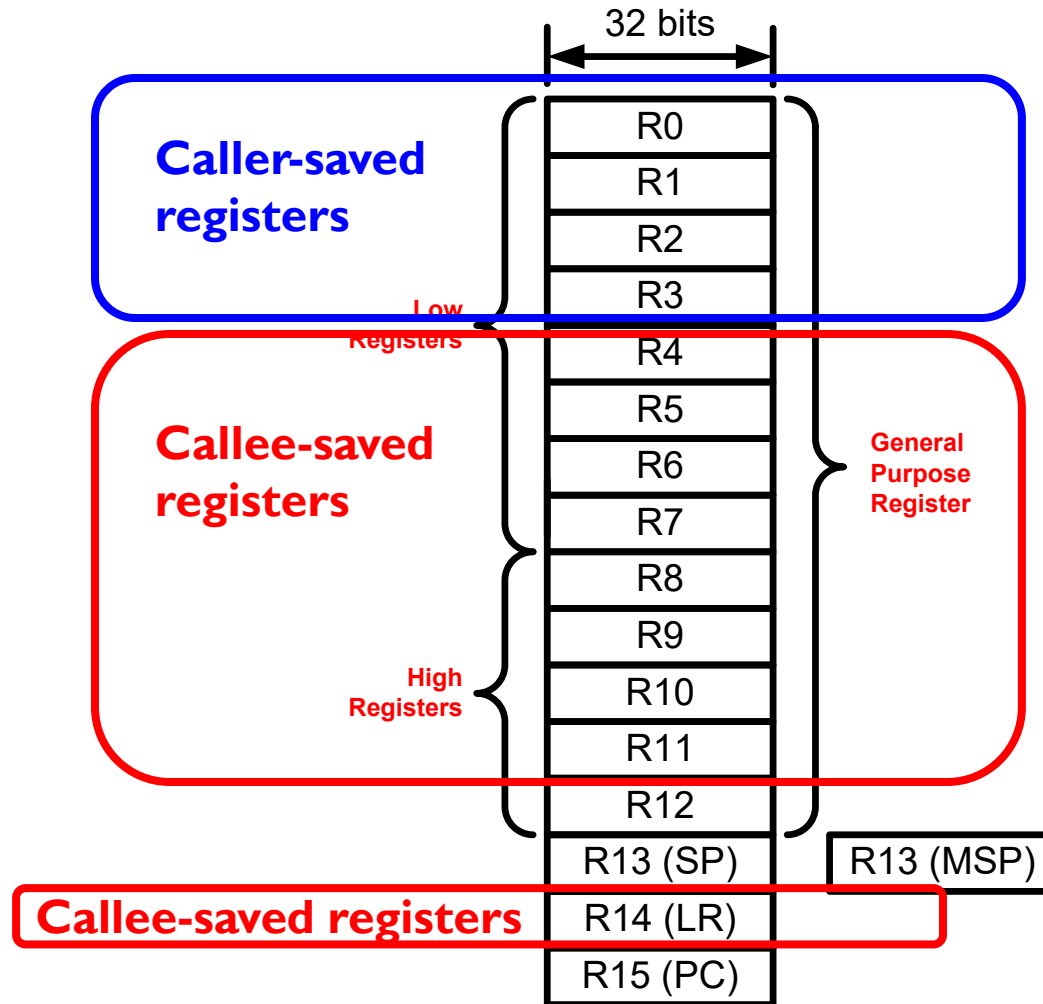
- ▶ $PC = LR$
 - ▶ Return to caller by setting PC to LR
- ▶ Equivalently:
 - ▶ **PUSH {LR}** at start of procedure
 - ▶ **POP {PC}** at end of procedure

Caller Program	Subroutine/Callee	
... BL foo ...	foo PROC ... BX LR EDP	foo PROC PUSH {LR} ... POP {PC} ; pops LR into PC (returns) EDP

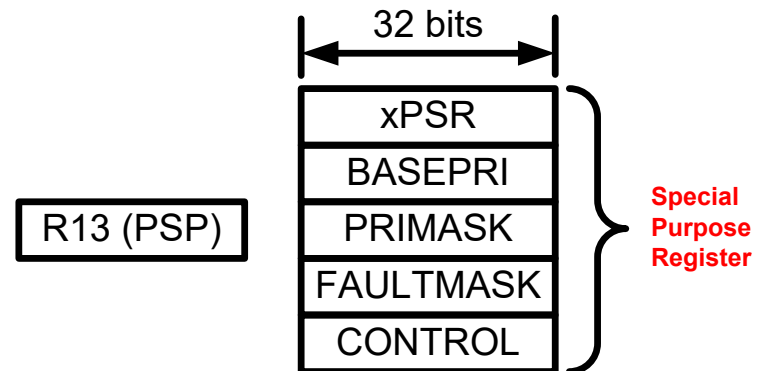
ARM Procedure Call Standard

Register	Usage	Subroutine Preserved	Notes
r0	Argument 1 and return value	No	If return has 64 bits, then r0:r1 hold it. If argument 1 has 64 bits, r0:r1 hold it.
r1	Argument 2	No	
r2	Argument 3	No	If the return has 128 bits, r0-r3 hold it.
r3	Argument 4	No	If more than 4 arguments, use the stack
r4	General-purpose V1	Yes	Variable register 1 holds a local variable.
r5	General-purpose V2	Yes	Variable register 2 holds a local variable.
r6	General-purpose V3	Yes	Variable register 3 holds a local variable.
r7	General-purpose V4	Yes	Variable register 4 holds a local variable.
r8	General-purpose V5	YES	Variable register 5 holds a local variable.
r9	Platform specific/V6	Yes/No	Usage is platform-dependent.
r10	General-purpose V7	Yes	Variable register 7 holds a local variable.
r11	General-purpose V8	Yes	Variable register 8 holds a local variable.
r12 (IP)	Intra-procedure-call register	No	It holds intermediate values between a procedure and the sub-procedure it calls.
r13 (SP)	Stack pointer	Yes	SP has to be the same after a subroutine has completed.
r14 (LR)	Link register	No	LR does not have to contain the same value after a subroutine has completed.
r15 (PC)	Program counter	N/A	Do not directly change PC

Caller-saved Registers *vs* Callee-saved Registers



- Not saved by subroutine
- Hold arguments/result
- Caller expects their values are retained
- Callee must save and store it if callee modifies it



Preserve Runtime Environment via Stack

Caller Program	Subroutine/Callee
Caller should save these registers if callers needs to re-use their original values: • R0 – R3 • R12 • CPSR	Callee should Preserve • R4 – R11 • R14 (LR) • R13 (SP)

► Top code:

- Callee foo does MOV R4, #10 → this overwrites whatever the caller had in R4.
- foo does BX LR → returns to caller, and R4 remains 10 in the caller.

► Middle code:

- Caller expects callee does not modify r4.
- Callee should preserve r4 by saving and restores R4 by PUSH and POP on the stack

► Bottom code:

- Callee saves and restores R4 by PUSH and POP
- Callee uses PUSH{LR} and POP{PC}, equivalent to BX LR

Caller Program	Subroutine/Callee
<pre>MOV r4, #100 ... BL foo ... ADD r4, r4, #1 ; r4 = 11</pre>	<pre>foo PROC ... MOV r4, #10 ; foo changes r4 ... BX LR EDP</pre>
Caller Program	Subroutine/Callee
<pre>MOV r4, #100 ... BL foo ... ADD r4, r4, #1 ; r4 = 101</pre>	<pre>foo PROC PUSH {R4} ; save caller's R4 MOV R4, #10 POP {R4} ; restore caller's R4 BX LR EDP</pre>
Caller Program	Subroutine/Callee
<pre>MOV r4, #100 ... BL foo ... ADD r4, r4, #1 ; r4 = 101</pre>	<pre>foo PROC PUSH {R4, LR} MOV R4, #10 POP {R4, PC} ; pops LR into PC (returns) EDP</pre>

Nested Subroutines: What is wrong?

Caller Program	Subroutine foo	Subroutine bar
<pre>MOV r4, #100 ... BL foo ... ADD r4, r4, #1</pre>	<pre>foo PROC PUSH {r4} ... MOV r4, #10 ... BL bar ... POP {r4} BX LR ENDP</pre>	<pre>bar PROC ... BX LR ENDP</pre>

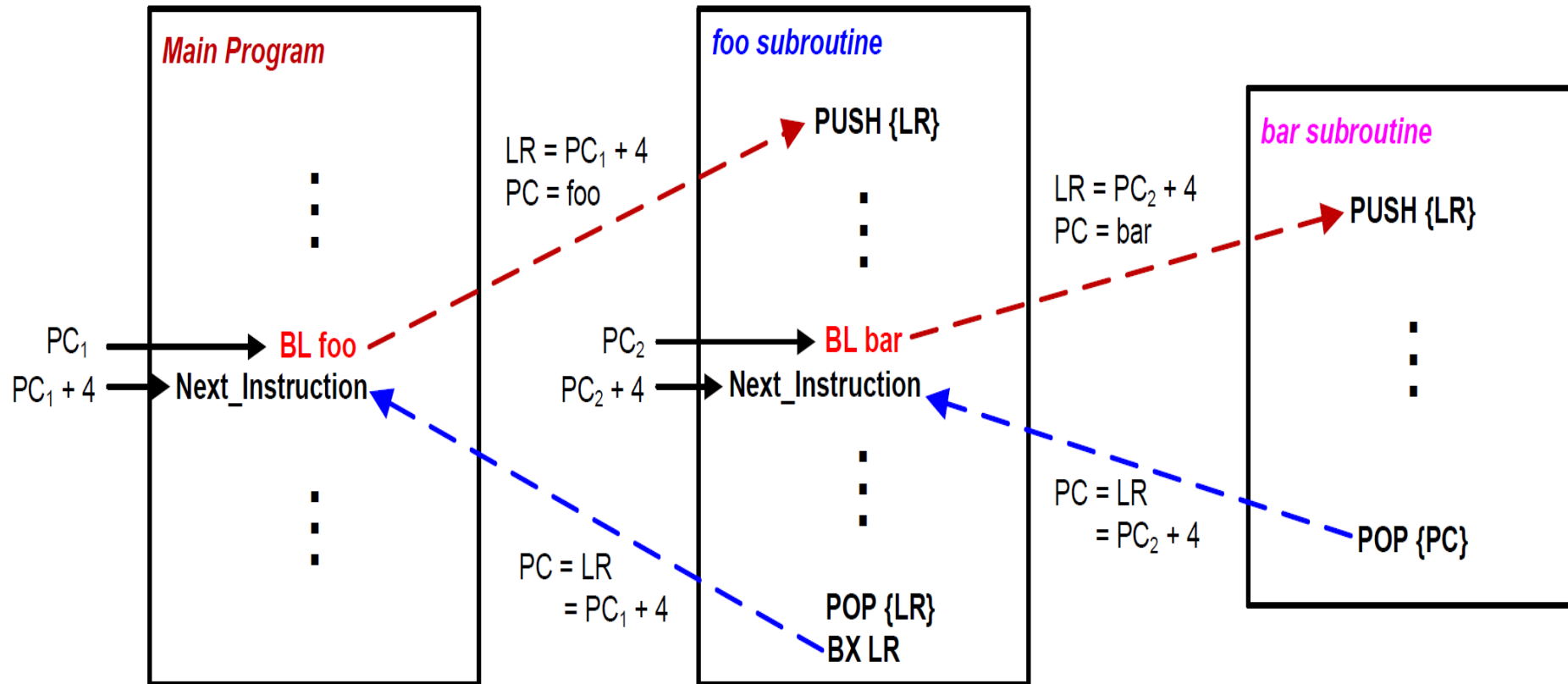
- ▶ Caller does BL foo → LR = return address back into the caller.
- ▶ foo does PUSH {r4} (saves r4) but does not save LR.
- ▶ foo does BL bar → this instruction overwrites LR with the return address back into foo (i.e. the instruction after BL bar).
- ▶ bar returns (BX LR) into foo (normal), but the original LR that pointed back to the caller was lost.
- ▶ foo does POP {r4} then BX LR — but LR now points to the instruction inside foo (not to the caller), so foo does not return to the caller

Nested Subroutines: Solution #1

foo saves and restores its LR for returning to its caller, before calling bar.

Caller Program	Subroutine foo	Subroutine bar
<pre>MOV r4, #100 ... BL foo ... ADD r4, r4, #1</pre>	<pre>foo PROC PUSH {r4, LR} ... MOV r4, #10 ... BL bar ... POP {r4, LR} BX LR ENDP</pre>	<pre>bar PROC ... BX LR ENDP</pre>

Nested Subroutines: Solution #1

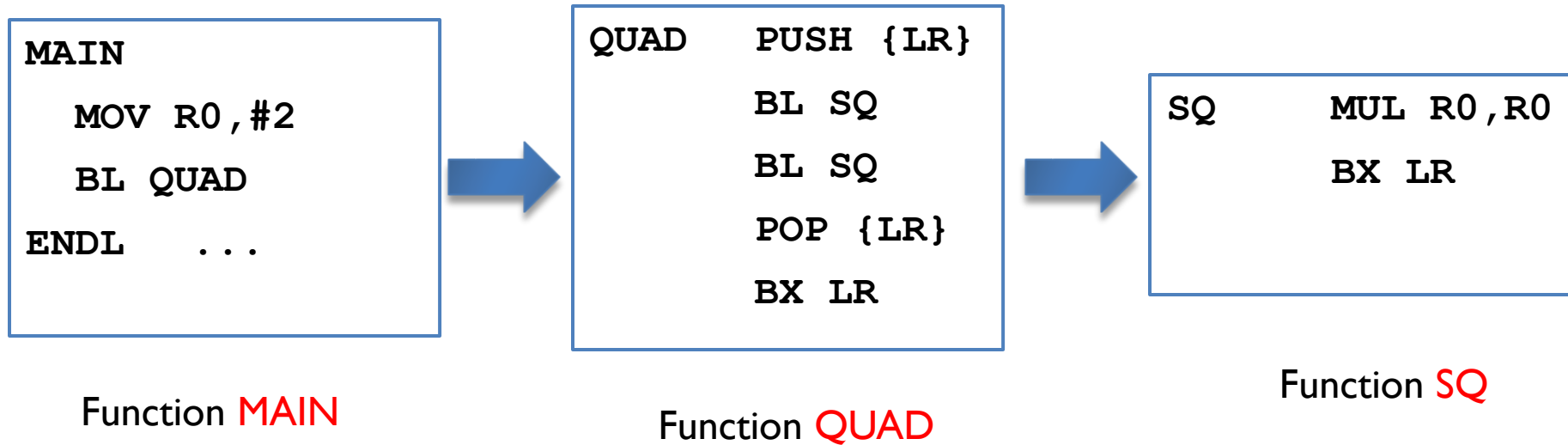


Nested Subroutines: Solution #2

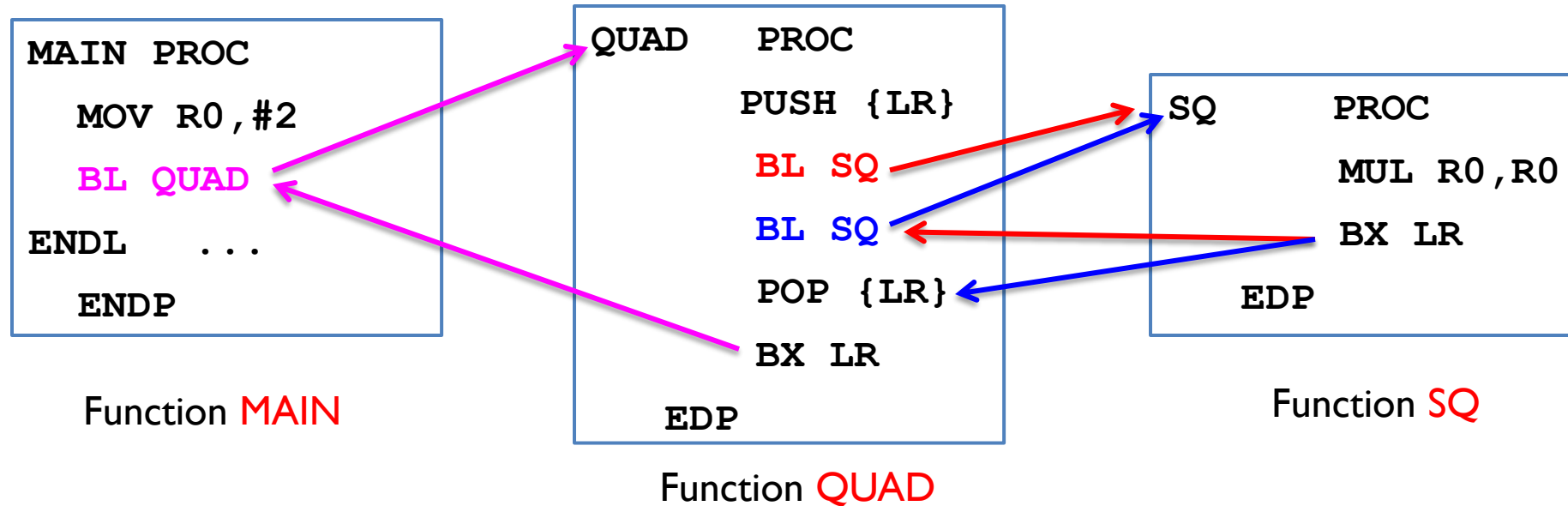
POP {r4, PC} is equivalent to POP {r4, LR} followed by BX LR.

Caller Program	Subroutine foo	Subroutine bar
<pre>MOV r4, #100 ... BL foo ... ADD r4, r4, #1</pre>	<pre>foo PROC PUSH {r4, LR} ... MOV r4, #10 ... BL bar ... POP {r4, PC} BX LR ENDP</pre>	<pre>bar PROC ... BX LR ENDP</pre>

Subroutine Calling Another Subroutine



Subroutine Calling Another Subroutine



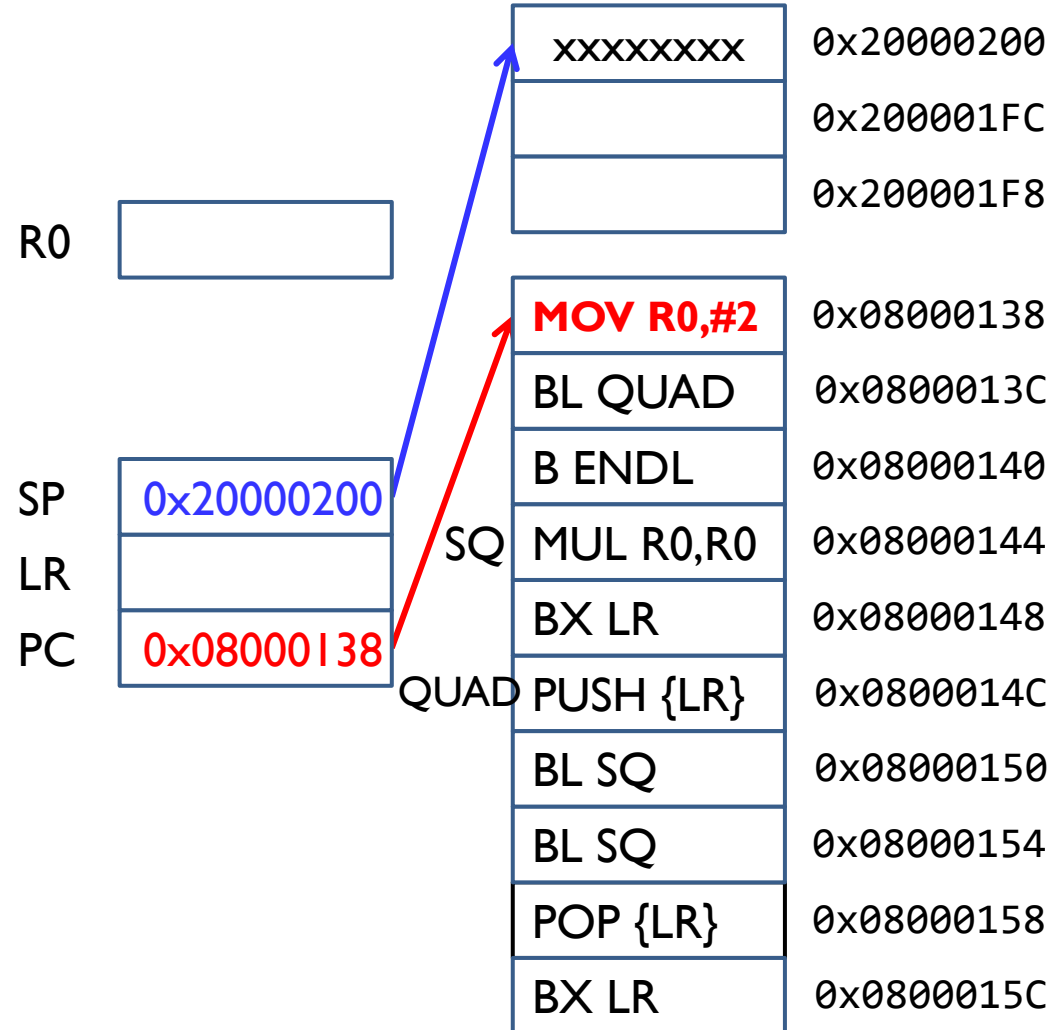
Example: $R0 = R0^4$

```
MOV R0,#2
BL QUAD
B ENDL

SQ    MUL R0,R0
      BX LR

QUAD  PUSH {LR}
      BL SQ
      BL SQ
      POP {LR}
      BX LR

ENDL  ...
```



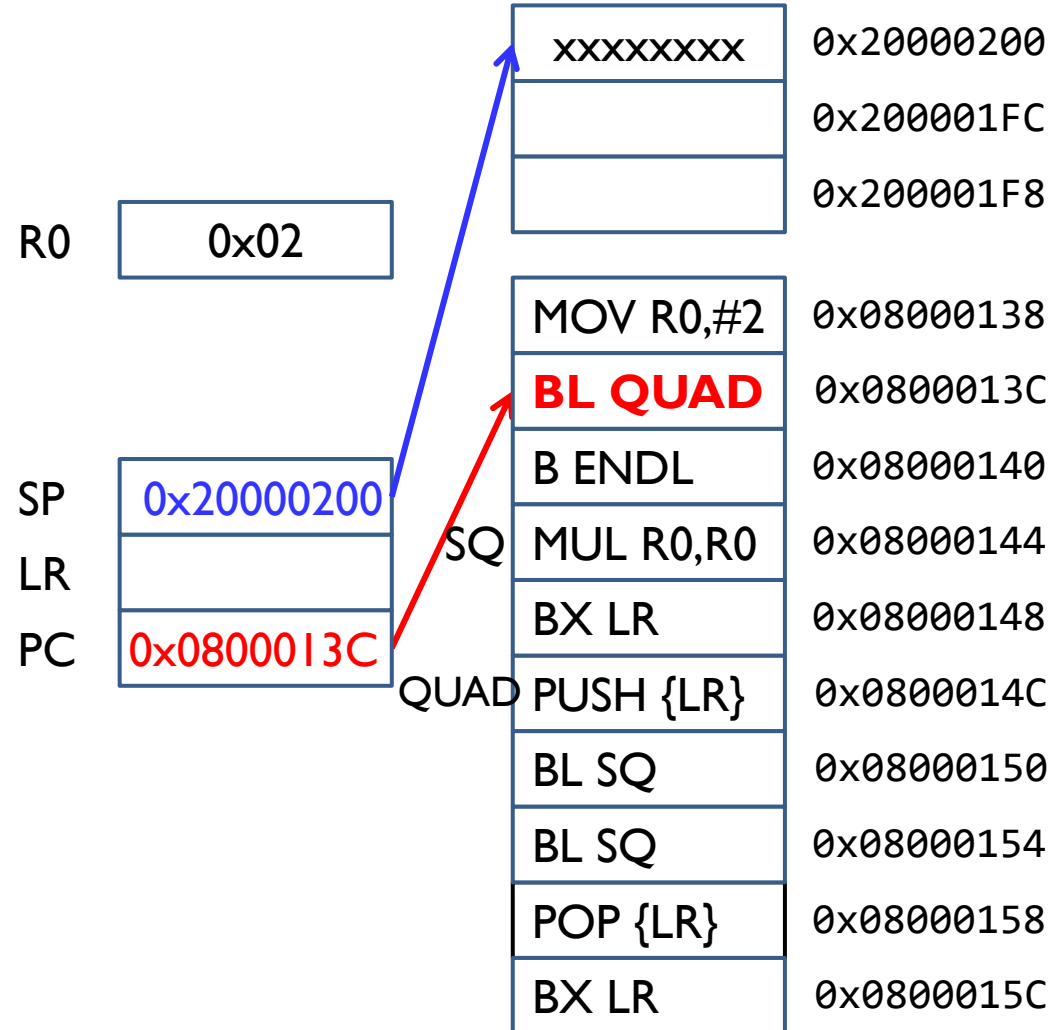
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ENDL  ...
```



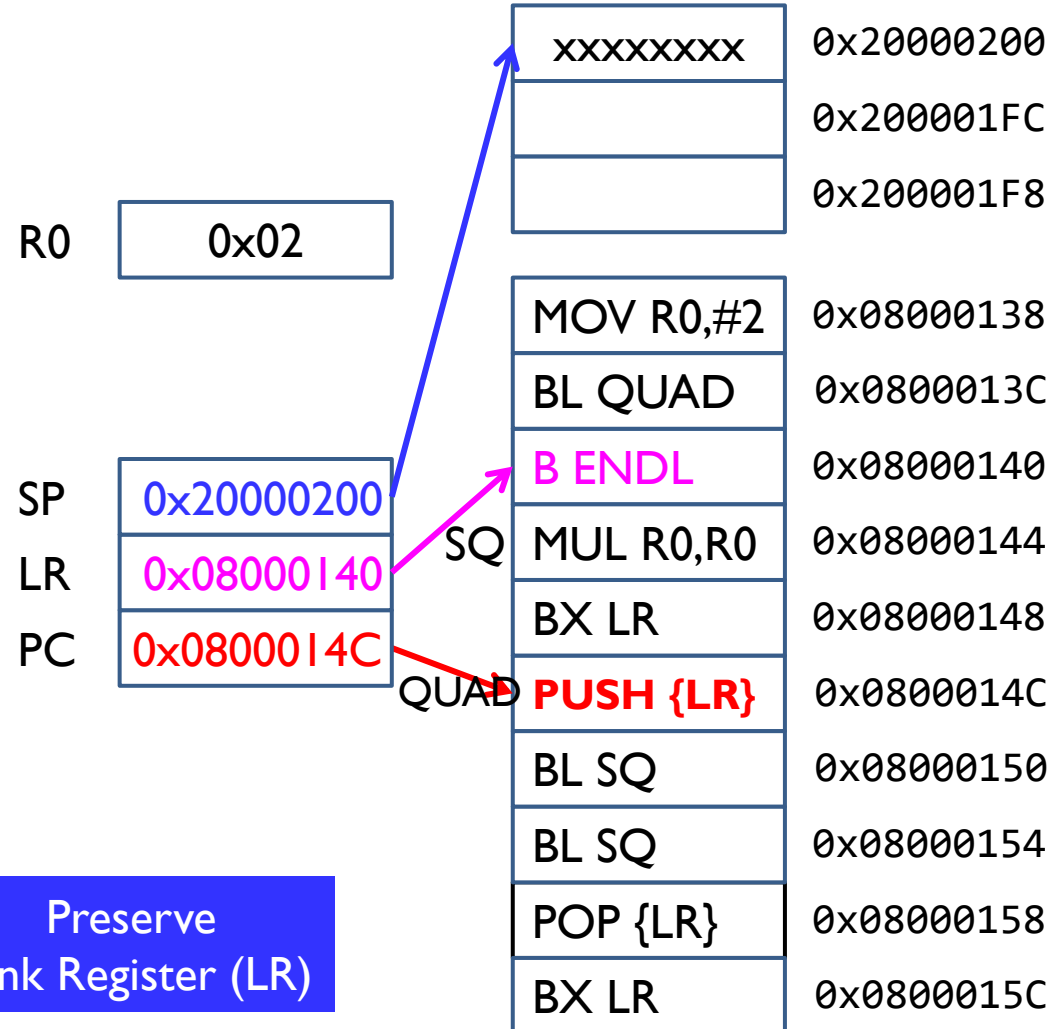
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ENDL  ...
```



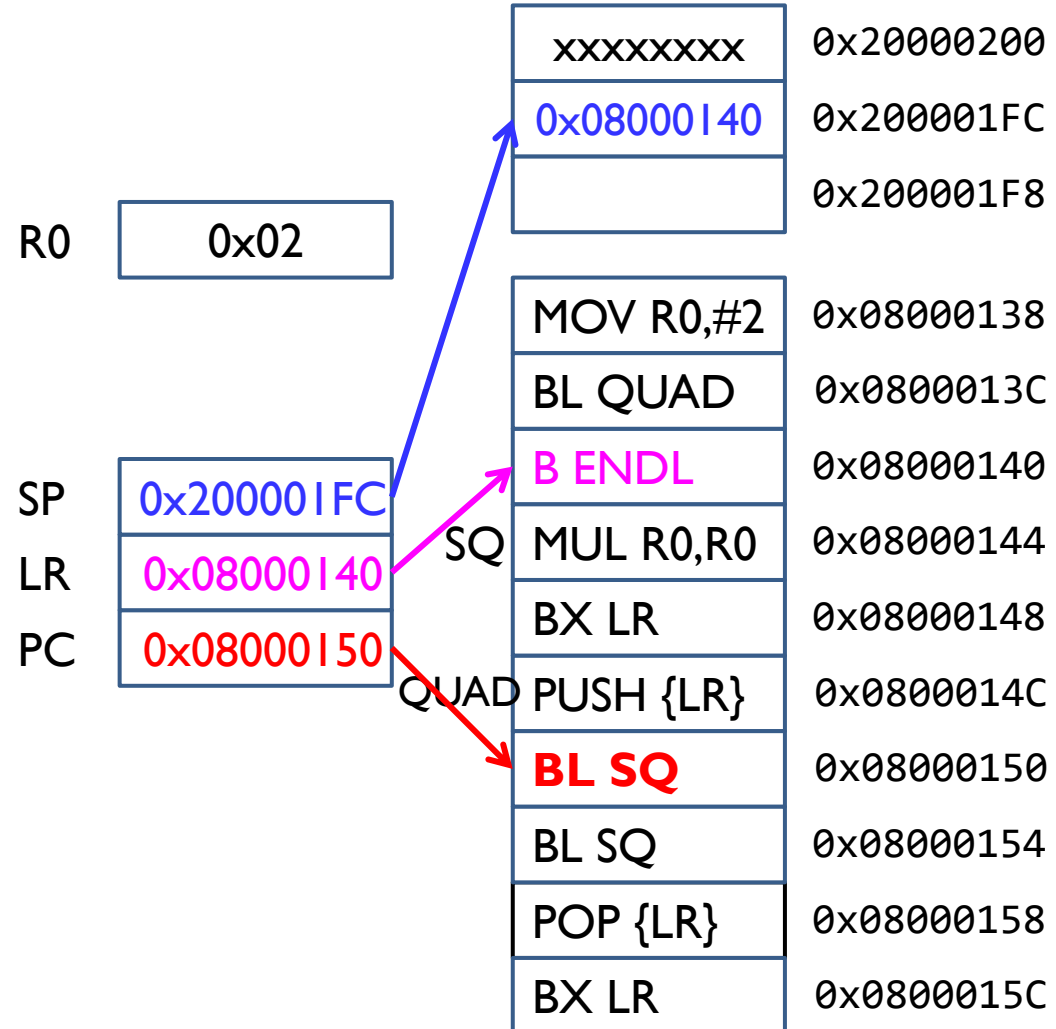
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ENDL  ...
```



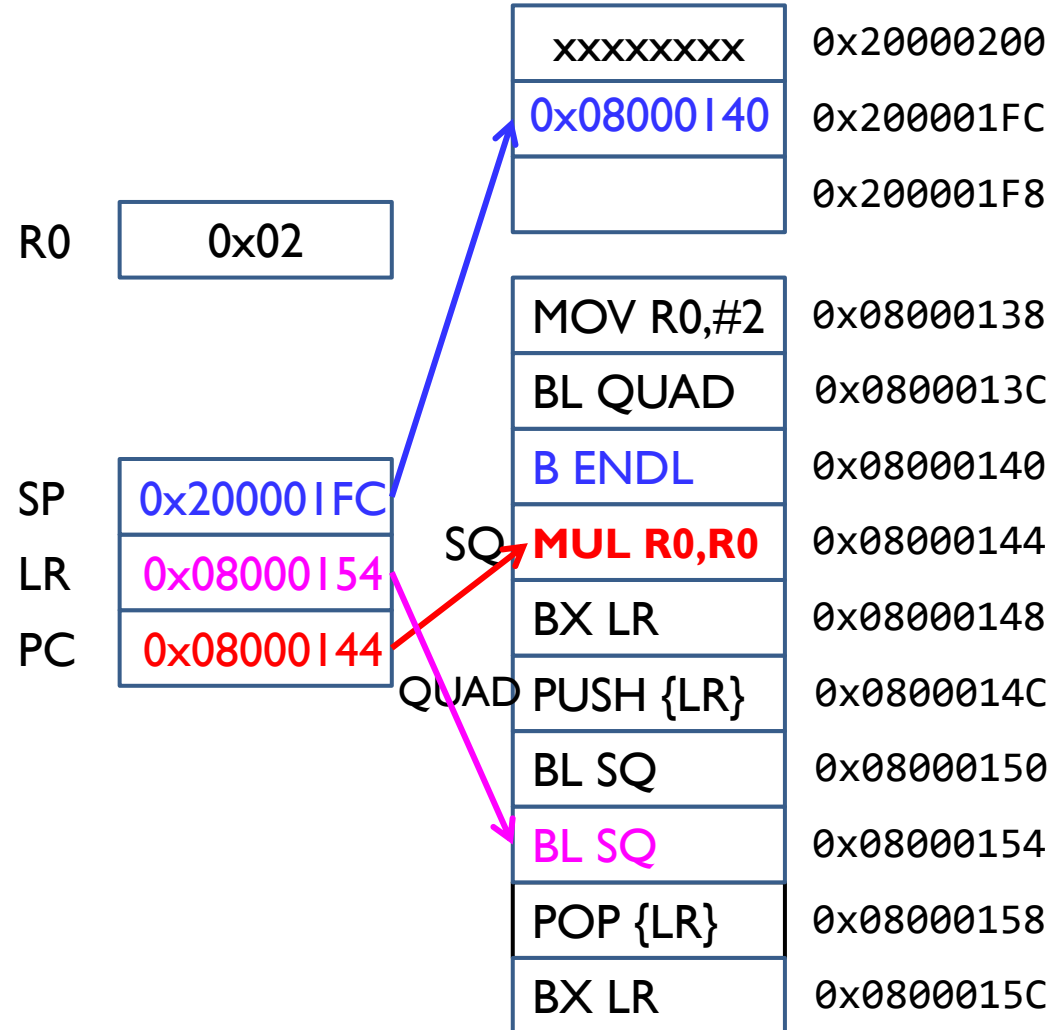
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SQ    MUL R0,R0
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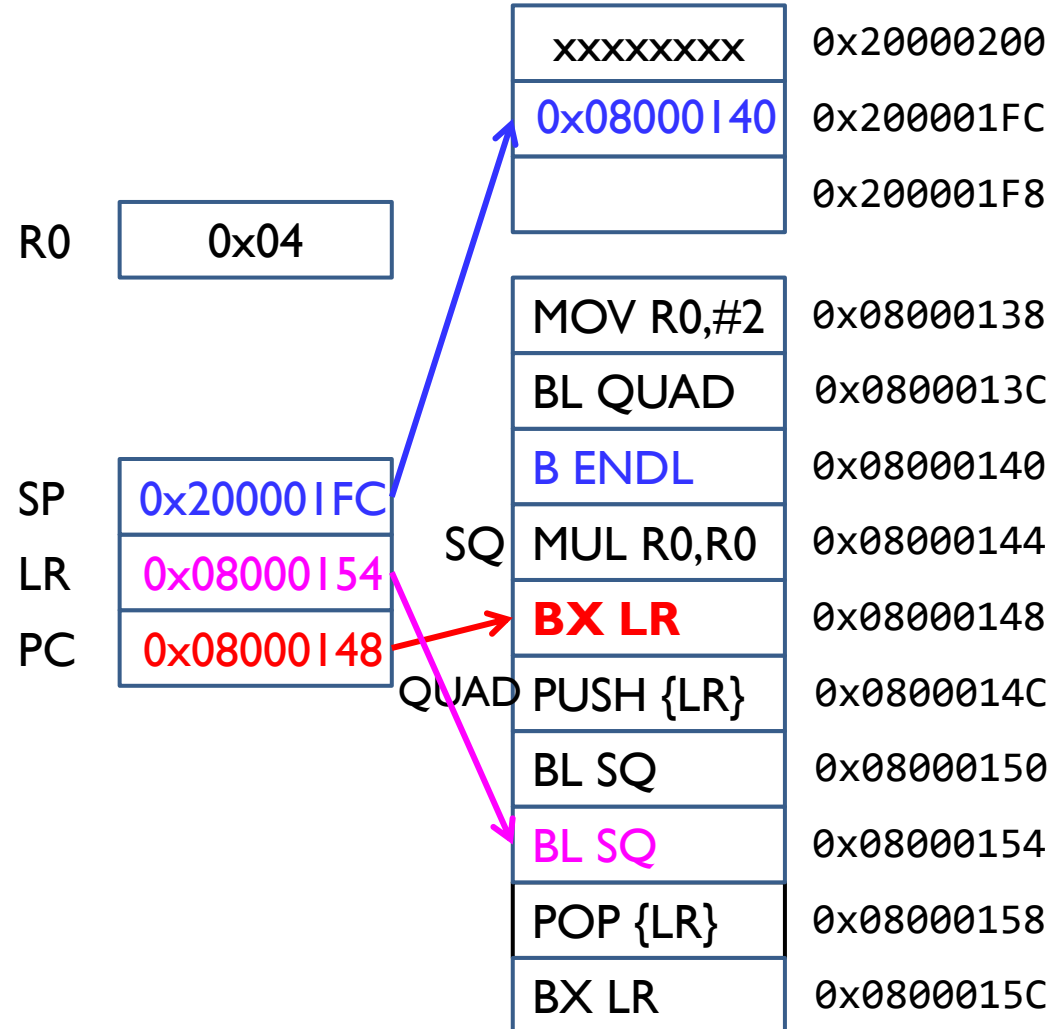
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      POP {LR}
      BX LR

ENDL  ...
```



Example: $R0 = R0^4$

	MOV R0,#2
	BL QUAD
	B ENDL
SQ	MUL R0,R0
	BX LR
QUAD	PUSH {LR}
	BL SQ
	BL SQ
	POP {LR}
	BX LR
ENDL	...



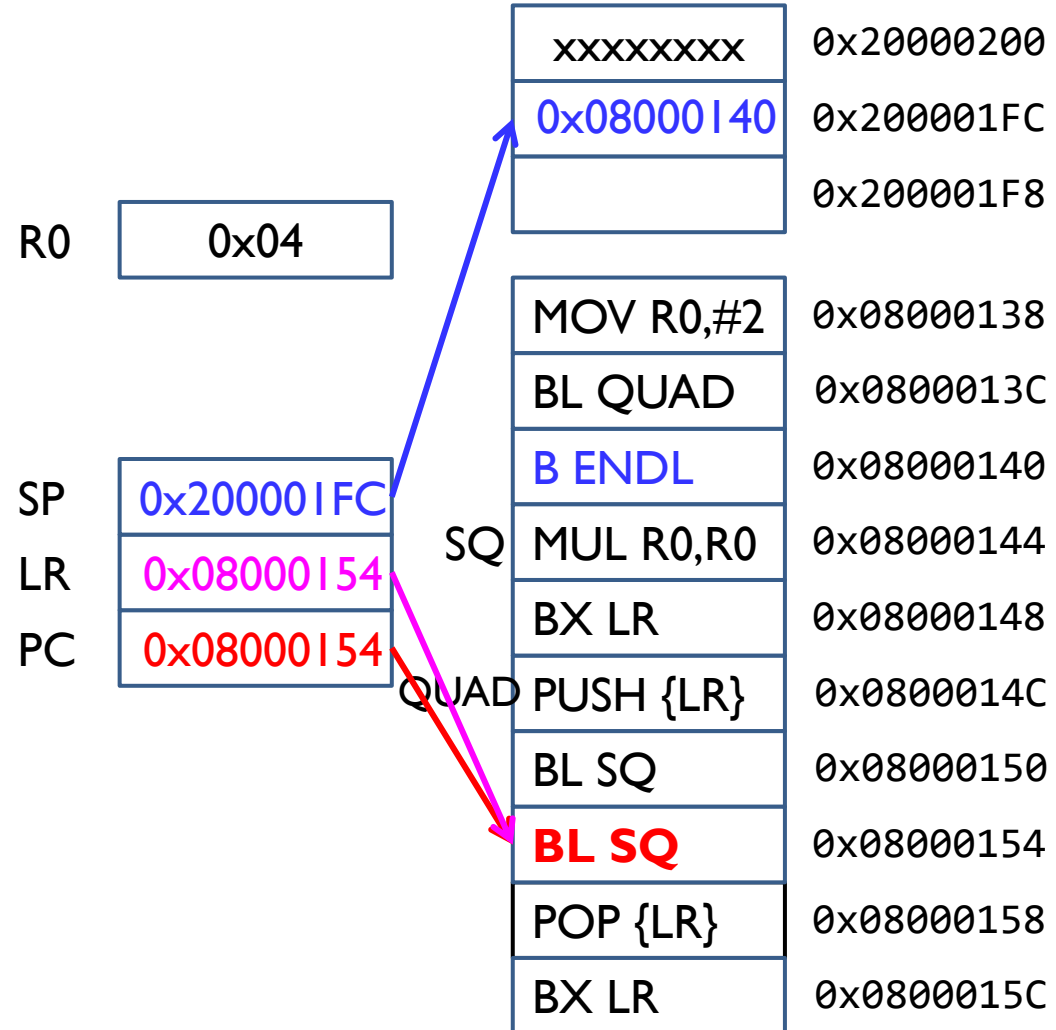
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SQ    MUL R0,R0
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QUAD  PUSH {LR}
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ENDL  . . .
```



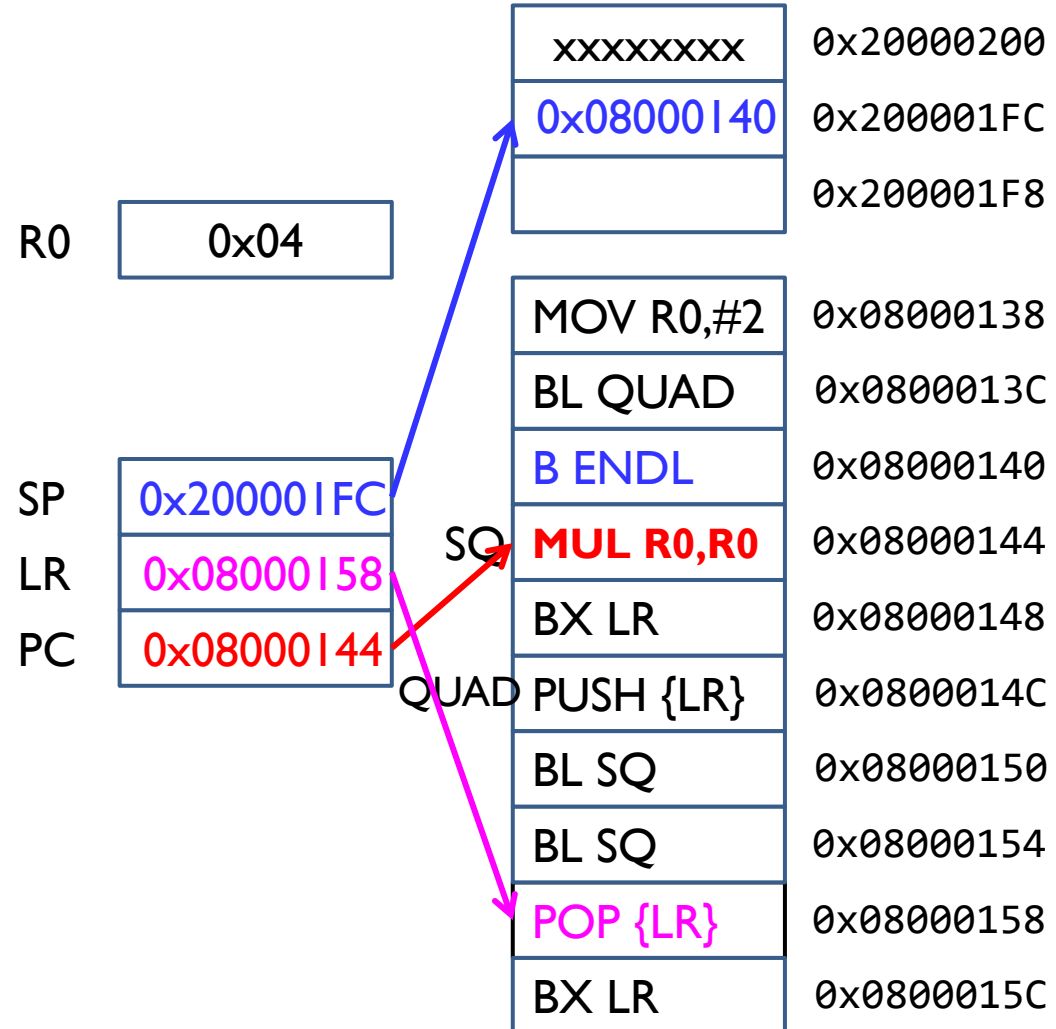
Example: $R0 = R0^4$

```
MOV R0,#2
BL QUAD
B ENDL

SQ    MUL R0,R0
BX LR

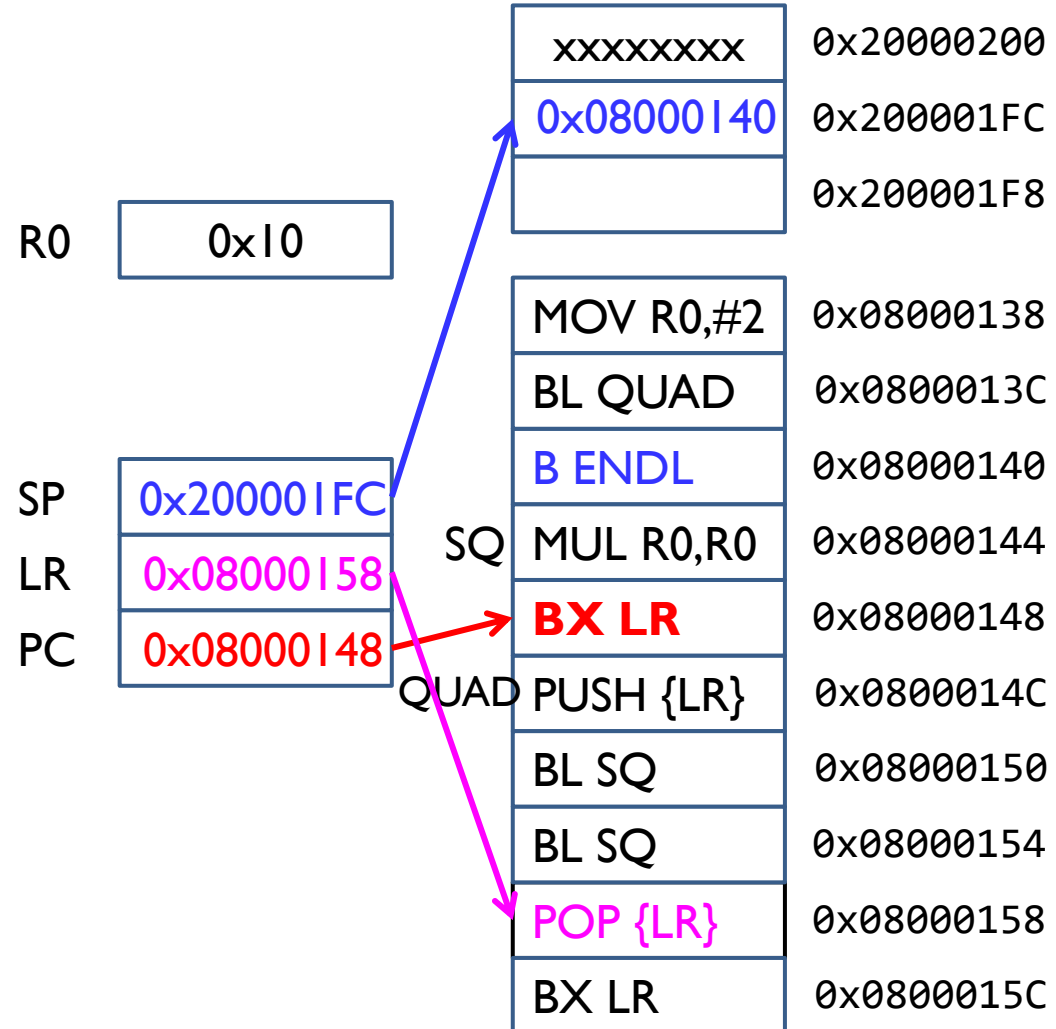
QUAD  PUSH {LR}
      BL SQ
      BL SQ
      POP {LR}
      BX LR

ENDL  ...
```



Example: $R0 = R0^4$

	MOV R0,#2
	BL QUAD
	B ENDL
SQ	MUL R0,R0
	BX LR
QUAD	PUSH {LR}
	BL SQ
	BL SQ
	POP {LR}
	BX LR
ENDL	...



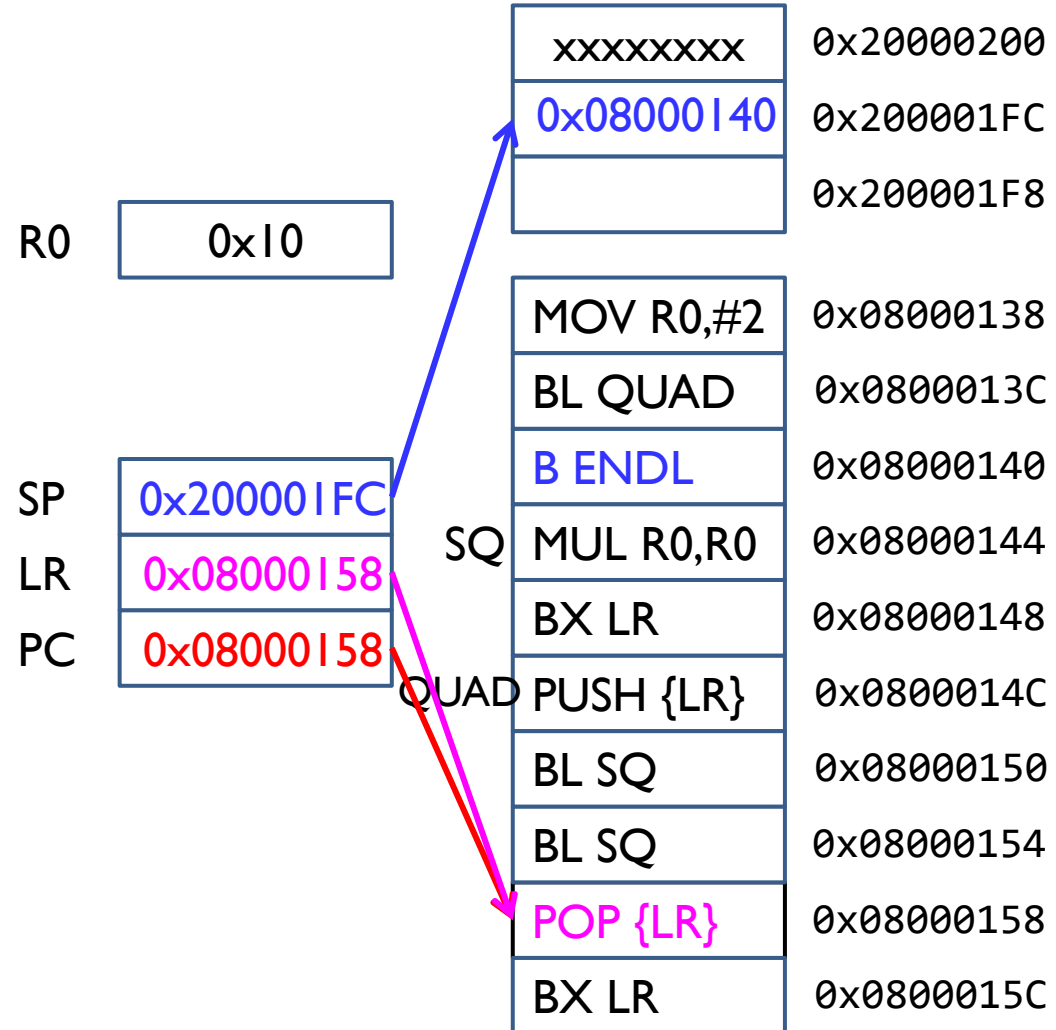
Example: $R0 = R0^4$

```
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BL QUAD
B ENDL

SQ    MUL R0,R0
      BX LR

QUAD  PUSH {LR}
      BL SQ
      BL SQ
      POP {LR}
      BX LR

ENDL  . . .
```



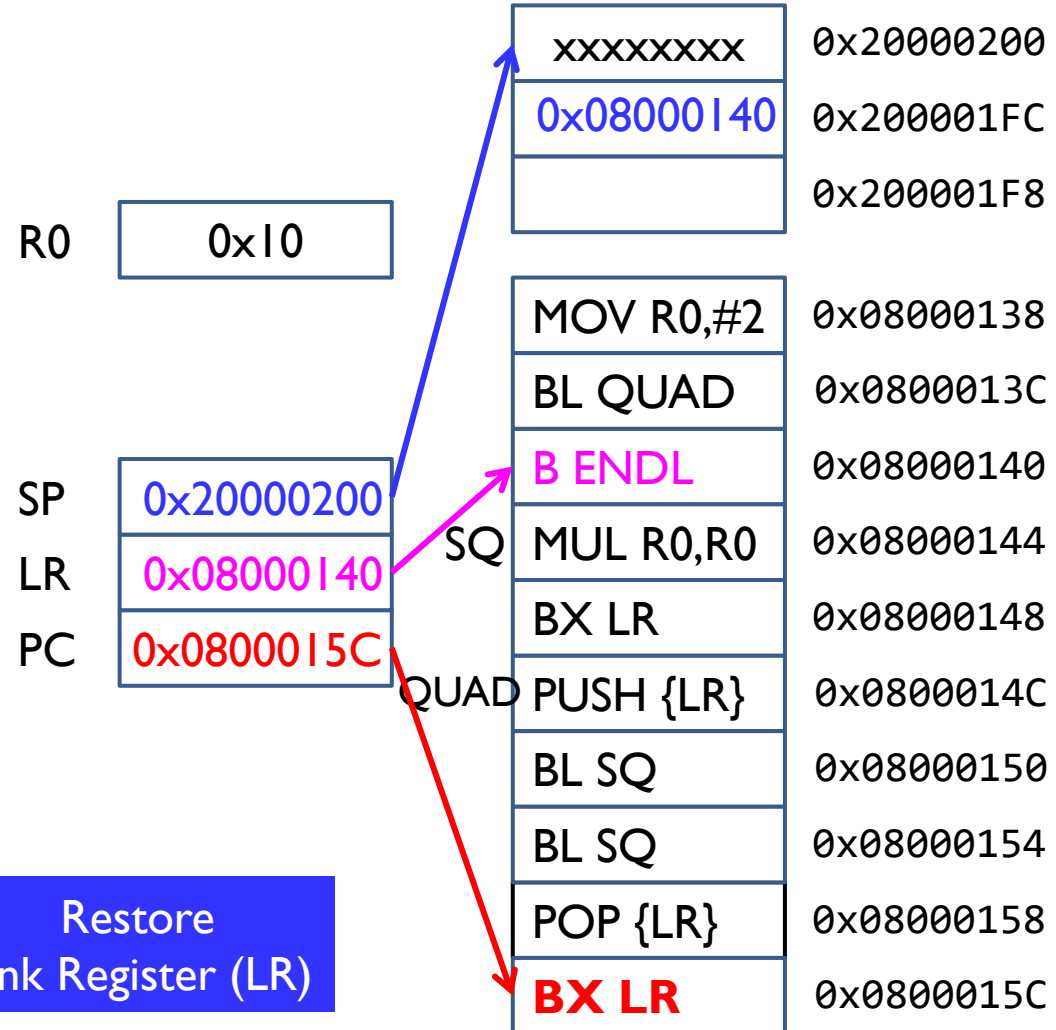
Example: $R0 = R0^4$

```
MOV R0,#2
BL QUAD
B ENDL

SQ    MUL R0,R0
      BX LR

QUAD  PUSH {LR}
      BL SQ
      BL SQ
      POP {LR}
      BX LR

ENDL  . . .
```



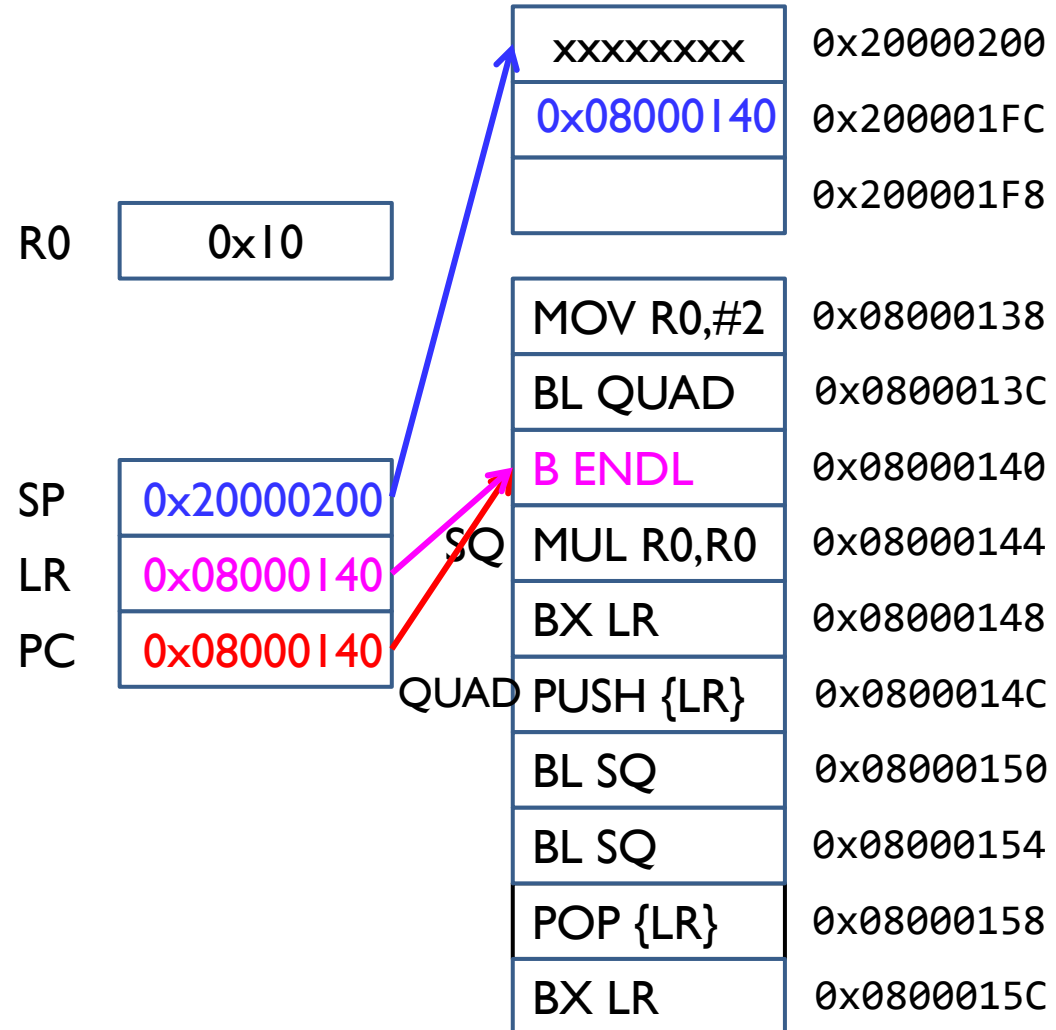
Example: $R0 = R0^4$

```
MOV R0,#2
BL QUAD
B ENDL

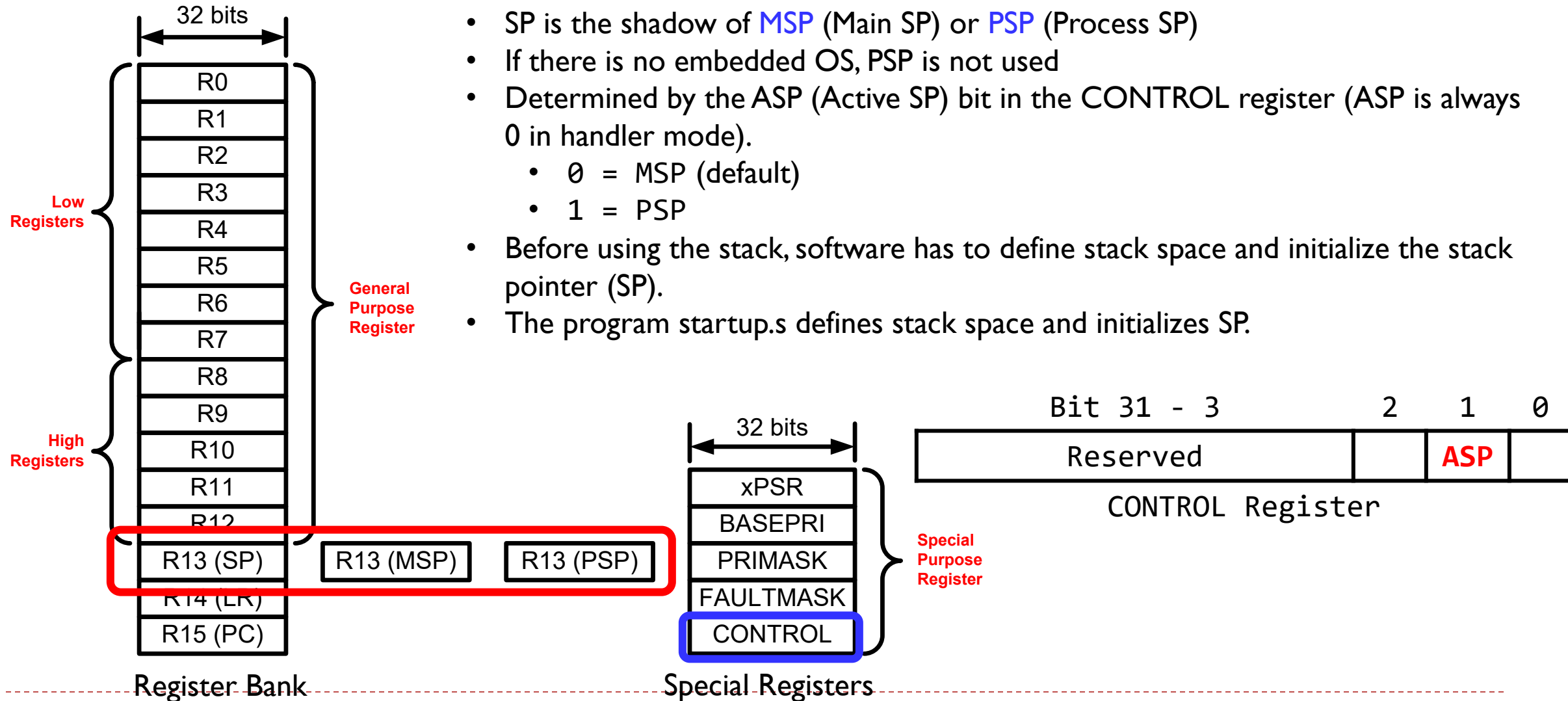
SQ    MUL R0,R0
      BX LR

QUAD  PUSH {LR}
      BL SQ
      BL SQ
      POP {LR}
      BX LR

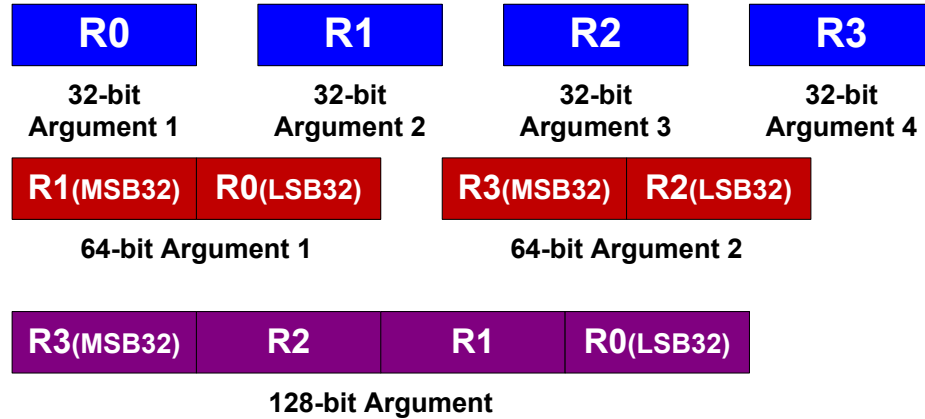
ENDL  ...
```



Stack Pointer (SP)



Passing Arguments into a Subroutine



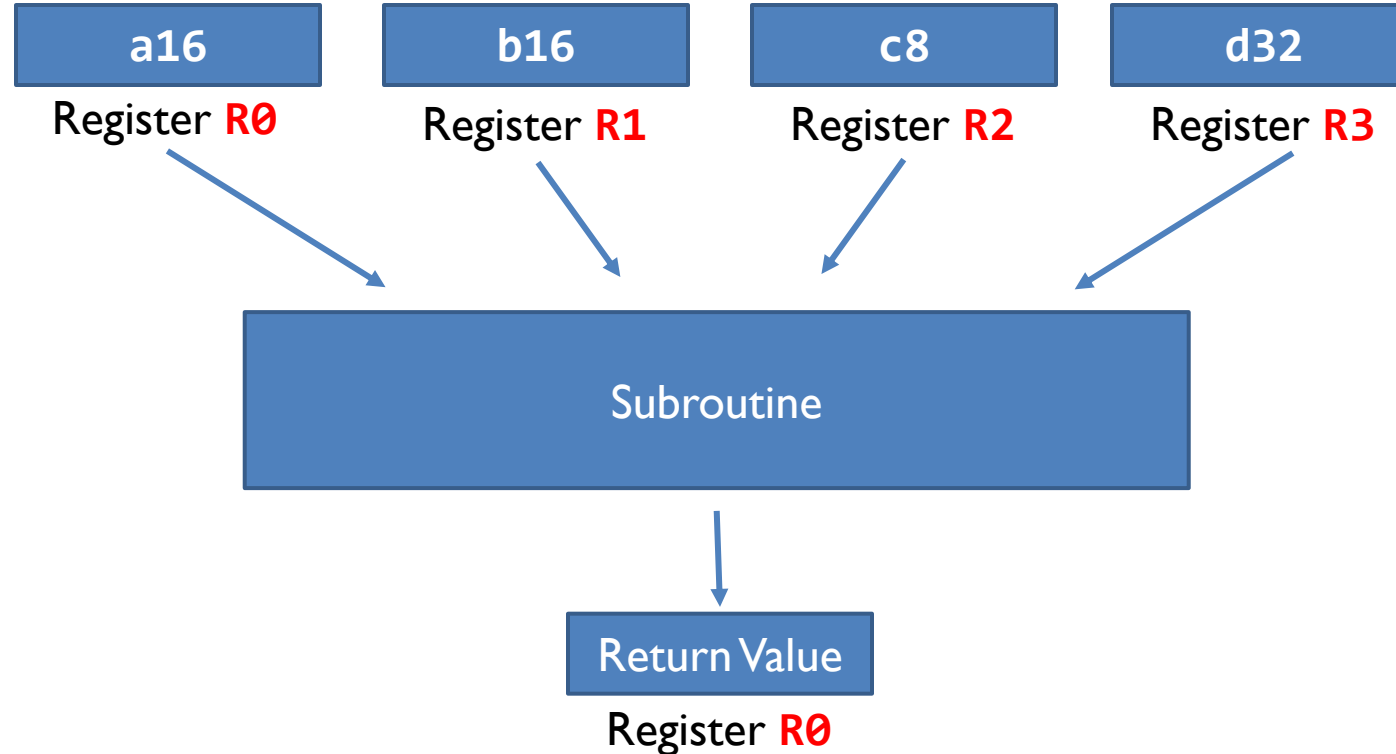
Extra arguments are pushed to the stack by the caller. The caller is responsible to pop them out of the stack after the subroutine returns.

Subroutine



Passing Arguments into a Subroutine

```
int32_t sum(int16_t a16, int16_t b16, int8_t c8, int32_t d32);
```



Passing 4 Arguments

```
int32_t sum(int16_t a16, int16_t b16, int8_t c8, int32_t d32);
```

```
s = sum(1, 2, 3, 4);
```

Caller

```
MOVS r0, #1 ; a16
MOVS r1, #2 ; b16
MOVS r2, #3 ; c8
MOVS r3, #4 ; d32
BL    sum
```

Callee

```
sum PROC
    ADD r0, r0, r1 ; a16 + b16
    ADD r0, r0, r2 ; add c8
    ADD r0, r0, r3 ; add d32
    BX  LR          ; return
ENDP
```

Passing Extra Arguments via Stack

Version 1

```
int32_t sum(int32_t a, int32_t b, int32_t c,  
int32_t d, int32_t h, int32_t i, int32_t j,  
int32_t k);  
s = sum(1, 2, 3, 4, 5, 6, 7, 8);
```

Caller

```
MOVS r0, #5  
MOVS r1, #6  
MOVS r2, #7  
MOVS r3, #8  
PUSH {r0, r1, r2, r3}  
MOVS r0, #1  
MOVS r1, #2  
MOVS r2, #3  
MOVS r3, #4  
BL    sum  
...  
POP {r0, r1, r2, r3}
```

Version 2

sum PROC

```
EXPORT sum  
ADD r0, r0, r1    ; add a + b  
ADD r0, r0, r2    ; add c  
ADD r0, r0, r3    ; add d  
LDRD r1,r2, [sp] ; r1=mem[sp],r2=mem[sp+4]  
ADD r0, r0, r1    ; add h  
ADD r0, r0, r2    ; add i  
LDRD r1,r2, [sp, #8] ; r1=mem[sp+8],r2=mem[sp+12]  
ADD r0, r0, r1    ; add j  
ADD r0, r0, r2    ; add k  
BX   LR  
ENDP
```

sum PROC

```
EXPORT sum  
PUSH {r5, r6, lr}  
ADD r0, r0, r1    ; add a + b  
ADD r0, r0, r2    ; add c  
ADD r0, r0, r3    ; add d  
LDRD r5,r6, [sp, #12] ; r5=mem[sp+12],r6=mem[sp+16]  
ADD r0, r0, r5    ; add h  
ADD r0, r0, r6    ; add i  
LDRD r5,r6, [sp, #20] ; r5=mem[sp+20],r6=mem[sp+24]  
ADD r0, r0, r5    ; add j  
ADD r0, r0, r6    ; add k  
POP {r5, r6, pc}  
ENDP
```

Explanations

► Version 1:

- Callee reads the extra args directly from the caller's push with LDRD: Load Register Doubleword:
 - `LDRD r1,r2, [sp]` → loads 5th & 6th args (h, i)
 - `LDRD r1,r2, [sp,#8]` → loads 7th & 8th args (j, k)
- It returns with BX LR.

► Version 2:

- Callee **pushes** r5, r6, and lr on entry: `PUSH {r5, r6, lr}`.
- Because it pushed, the extra-argument addresses are shifted, so it uses offsets like `[sp,#12]` and `[sp,#20]` to read the caller's arguments into r5,r6.
 - `LDRD r5,r6, [sp, #12]`
 - `LDRD r5,r6, [sp, #20]`
- It returns via `POP {r5, r6, pc}` (popping lr into pc returns directly).
- Ladder analogy:
 - Think of the stack like a ladder: The ladder = one shared stack in memory.
 - The rung each function stands on = its current SP value.
 - When a function “pushes,” it steps down a few rungs (SP decreases).
 - The caller's pushed data remains up above — just higher up the same ladder.

Address	Contents	Access
0x0FFC	arg8 (k=8)	[sp,#12]
0x0FF8	arg7 (j=7)	[sp,#8]
0x0FF4	arg6 (i=6)	[sp,#4]
0x0FF0	arg5 (h=5)	[sp] ← SP

Version 1 stack after Caller PUSH

Address	Contents	Access
0x0FFC	arg8 (k=8)	[sp,#24]
0x0FF8	arg7 (j=7)	[sp,#20]
0x0FF4	arg6 (i=6)	[sp,#16]
0x0FF0	arg5 (h=5)	[sp,#12]
0x0FEC	saved lr	[sp,#8]
0x0FE8	saved r6	[sp,#4]
0x0FE4	saved r5	[sp] ← SP

Version 2 stack after Caller PUSH and Callee PUSH

Version 2 is Better Programming Practice

- ▶ Caller pushes extra arguments (5th to 8th) onto the stack before calling the function.
- ▶ Version 1: Callee accesses those extra arguments directly from the stack using LDRD instructions at specific offsets from SP. It demonstrates the basic mechanism for handling extra arguments on the stack.
- ▶ Version 2: Callee begins by pushing registers r5, r6, and the link register (LR) onto the stack to preserve them. After addition operations, it pops r5, r6, and the program counter (PC) to return, restoring the preserved registers and the return address. This ensures that the subroutine does not unintentionally overwrite or lose the caller's data and return address, maintaining program correctness during and after the function call.

Summary

- ▶ ARM Cortex-M uses full descending stack
- ▶ How to pass arguments into a subroutine?
 - ▶ Each 8-, 16- or 32-bit variables is passed via r0, r1, r2, r3
 - ▶ Extra parameters are passed via stack
- ▶ What registers should be preserved?
 - ▶ Caller-saved registers vs callee-saved registers
- ▶ How to preserve the running environment for the caller?
 - ▶ Via stack

References

- ▶ Lecture 31. Preserving registers in a Subroutine
 - ▶ <https://www.youtube.com/watch?v=DGKjFKjxAYs&list=PLRJhV4hUhlymmp5CCelFPyxbknsdcXCc8&index=31>